

45. Quad Serial Peripheral Interface (QSPI)

This MCU includes a one-channel quad serial peripheral interface.

In this section, “PCLK” is used to refer to PCLKB.

45.1 Overview

Table 45.1 lists the specifications of the QSPI, Figure 45.1 shows a block diagram of the QSPI, and Table 45.2 lists the I/O pins of the QSPI.

Table 45.1 QSPI Specifications

Item	Description
Number of channels	1 channel
QSPI transfer function	Capable of communications to the serial flash memory through single-/dual-/quad-SPI operation - Single-SPI operation (full-duplex communications) - Use of QMO (master out), QMI (master in), QSSL (slave select), and QSPCLK (SPI clock) signals allow for communications to the serial flash memory through SPI operation (four-wire method). - QMO output pin and QMI input pin - QSSL and QSPCLK serve as output pins. - Dual-SPI operation (half-duplex communications) - Use of QIO1, QIO0, QSSL, and QSPCLK signals allow for communications to the serial flash memory through SPI operation (four-wire method). - QSSL and QSPCLK serve as output pins. - Bidirectional QIO1 and QIO0 pins - Quad-SPI operation (half-duplex communications) - Use of QIO3 to QIO0, QSSL, and QSPCLK signals allow for communications to the serial flash memory through SPI operation (six-wire method). - QSSL and QSPCLK serve as output pins. - Bidirectional QIO3 to QIO0 pins
Data format	- Transfer data length is selectable from 8 bits to 128 Gbits. - Data is continuously transferred 1 through 4,294,967,296 times in 8-, 16-, or 32-bit units.
Bit rate	- QSPCLK can be divided by 2 to 4080 (max.). - QSPCLK can be generated by dividing PCLK by the on-chip baud rate generator.
Buffer configuration	32 bytes for transmission 32 bytes for reception
Shift registers	32 bits each for transmission and reception
QSSL pin control	- Controllable delay from QSSL signal assertion to QSPCLK operation (clock delay) Range: 0 and 1.5 to 8.5 QSPCLK cycles (set in QSPCLK-cycle units) - Controllable delay from QSPCLK stoppage to QSSL output negation (QSSL negation delay) Range: 0 to 8 QSPCLK cycles (set in QSPCLK-cycle units) - Controllable wait for next-access QSSL output assertion (next-access delay) Range: 0 to 8 QSPCLK cycles (set in QSPCLK-cycle units) - Capable of holding QSSL output value from transfer end to next access - QSSL polarity can be changed.
Transfer control	- A transfer of up to four commands can be executed sequentially in looped execution. - Single-SPI transfer start condition: The specified length of data is in the transmit buffer and there is a space for the specified length of data in the receive buffer. However, when the SPDCR.TXDMY bit is set, receive operation can be started even when the specified length of data is not in the transmit buffer. - Dual-/quad-SPI write start condition: The specified length of data is in the transmit buffer. - Dual-/quad-SPI read start condition: There is a space for the specified length of data in the receive buffer. - QIO3 to QIO0 and QMO output values during the QSSL negation period can be specified. - QIO3 and QIO2 can be specified to output fixed values in single-/dual-SPI modes.
Interrupt sources	- Receive buffer full interrupt - Transmit buffer empty interrupt - QSSL negation interrupt
Others	- QSPI initialization function - Loop back mode
Low power consumption function	Module stop state can be set.

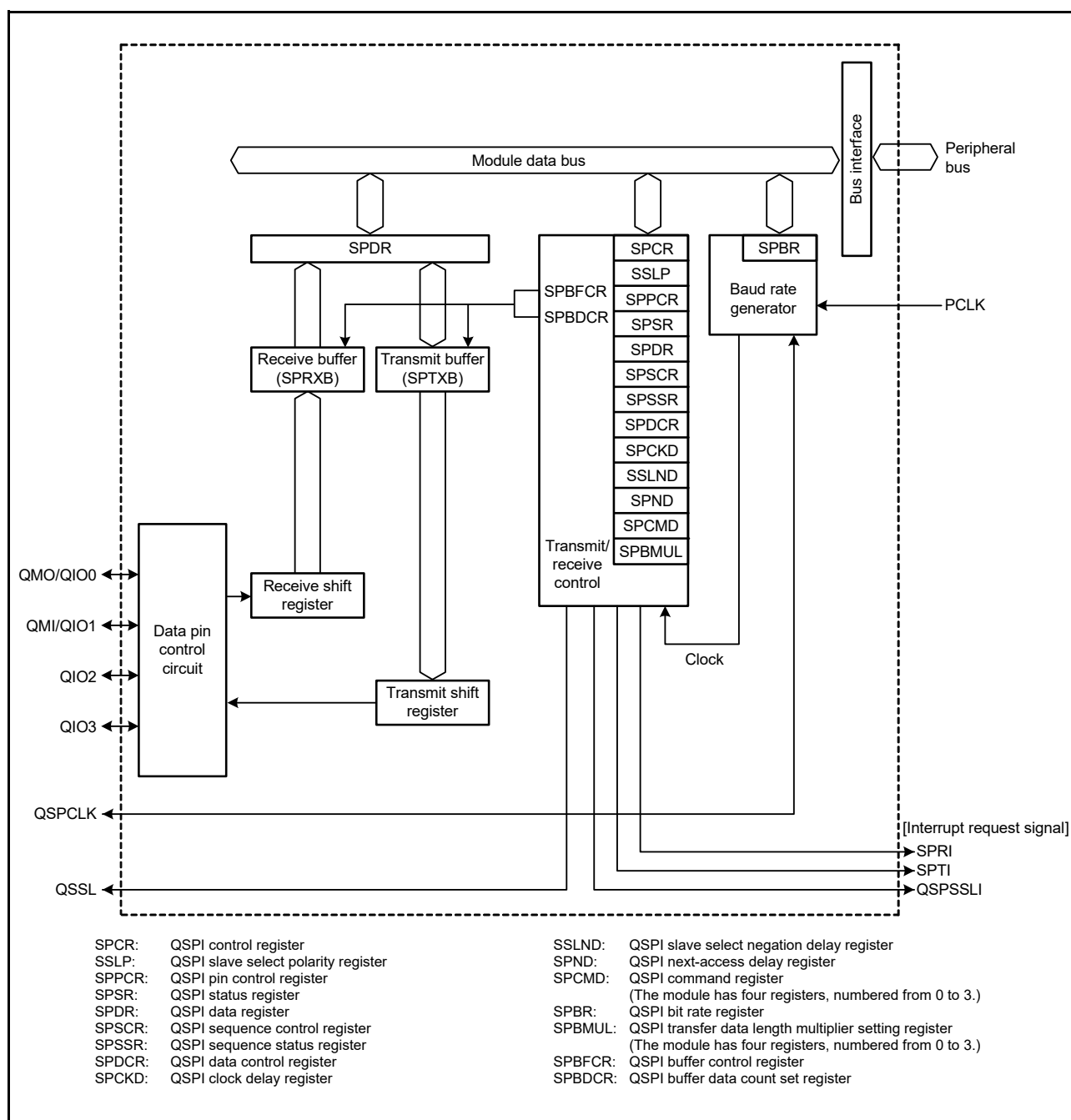


Figure 45.1 QSPI Block Diagram

Table 45.2 QSPI I/O Pins

Pin Name	I/O	Function
QSPCLK	Output	Clock output
QMO/QIO0*1	I/O	Master output data/data 0
QMI/QIO1*1	I/O	Master input data/data 1
QIO2*2	I/O	Data 2
QIO3*2	I/O	Data 3
QSSL	Output	Slave selection

Note 1. In single-SPI mode, QMO and QMI are enabled; QIO0 and QIO1 in dual-/quad-SPI modes.

Note 2. In single-/dual-SPI modes, fixed value according to register setting is output; QIO2 and QIO3 in quad-SPI mode.

45.2 Register Descriptions

45.2.1 QSPI Control Register (SPCR)

Address(es): QSPI.SPCR 0008 9E00h

b7	b6	b5	b4	b3	b2	b1	b0
SPRIE	SPE	SPTIE	—	MSTR	—	SPSSLIE	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	SPSSLIE	QSSL Negation Interrupt Enable	0: Disables interrupt requests. 1: Enables interrupt requests.	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	MSTR	QSPI Master/Slave Mode Select *1	0: Slave mode 1: Master mode (output from QSPCLK and QSSL pins enabled)	R/W
b4	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5	SPTIE	Transmit Interrupt Enable	0: Disables interrupt requests. 1: Enables interrupt requests.	R/W
b6	SPE	SPI Function Enable	0: Disables the function. 1: Enables the function.	R/W
b7	SPRIE	Receive Interrupt Enable	0: Disables interrupt requests. 1: Enables interrupt requests.	R/W

Note 1. This MCU does not support slave mode. When using this bit, set it to 1.

The SPCR register sets the operating mode for the QSPI. If the set value of the MSTR bit is changed while the SPE bit is 1, subsequent operation cannot be guaranteed.

SPSSLIE Bit (QSSL Negation Interrupt Enable)

This bit enables or disables generation of QSSL negation interrupt requests when the SPSR.SPSSLF flag is set to 1.

MSTR Bit (QSPI Master/Slave Mode Select)

This bit selects master or slave mode. Setting this bit to 1 allows the QSPCLK and QSSL pins to be used as output pins. When setting the MSTR bit, specify the following while the SPE bit is 0, and then set the MSTR bit to 1.

- SSLP.SSLP bit setting (QSSL pin level setting)
- SPCMDn.CPOL bit (n = 0 to 3) setting (QSPCLK pin level setting)

Note that this MCU does not support slave mode.

SPTIE Bit (Transmit Interrupt Enable)

This bit enables or disables generation of transmit interrupt requests when the number of transmit data units in the transmit buffer is equal to or less than the specified transmit buffer data triggering number leading the corresponding flag in the SPSR register being set to 1.

SPE Bit (SPI Function Enable)

Setting this bit to 1 enables the module function.

Setting this bit to 0 disables and partially initializes the module function.

SPRIE Bit (Receive Interrupt Enable)

This bit enables or disables generation of receive interrupt requests when the number of receive data units in the receive buffer is equal to or greater than the specified receive buffer data triggering number leading the corresponding flag in the status register being set to 1.

45.2.2 QSPI Slave Select Polarity Register (SSLP)

Address(es): QSPI.SSLP 0008 9E01h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	SSLP
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SSLP	QSSL Signal Polarity Setting	0: QSSL signal is active-low. 1: QSSL signal is active-high.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SSLP register sets the polarity of the QSSL signal. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

SSLP Bit (QSSL Signal Polarity Setting)

This bit sets the polarity of the QSSL signal. The polarity should be same as that of the device to be communicated with.

45.2.3 QSPI Pin Control Register (SPPCR)

Address(es): QSPI.SPPCR 0008 9E02h

b7	b6	b5	b4	b3	b2	b1	b0	
—	—	MOIFE	MOIFV	—	IO3FV	IO2FV	SPLP	
Value after reset:	0	0	0	0	0	1	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	SPLP	Loopback mode*1	0: Normal mode 1: Loopback mode	R/W
b1	IO2FV	Single-/Dual-SPI Mode QIO2 Output Fixed Value	0: QIO2 output fixed value equals 0. 1: QIO2 output fixed value equals 1.	R/W
b2	IO3FV	Single-/Dual-SPI Mode QIO3 Output Fixed Value	0: QIO3 output fixed value equals 0. 1: QIO3 output fixed value equals 1.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	MOIFV	Idle State Output Data Fixed Value	0: Data output value in the idle state (during the QSSL negation period) is 0. 1: Data output value in the idle state (during the QSSL negation period) is 1.	R/W
b5	MOIFE	Idle State Output Data Fixing Enable *2	0: Output value equals final data from previous transfer. 1: Output value equals the value set in the MOIFV bit.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When the loopback mode is specified in dual-/quad-SPI modes, the SPCMDn.SPRW bit (n = 0 to 3) should be set to 0 (write operation).

Note 2. In dual-/quad-SPI modes, QIO1 and QIO0/QIO3 to QIO0 are driven to the Hi-Z state after read transfer regardless of this bit setting (see section 45.3.2, Pin Control).

The SPPCR register sets the modes of the pins. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

SPLP Bit (Loopback mode)

When the SPLP bit is set to 1, this module shuts off the path between the data I/O pin and the transmit/receive shift register, and connects the input path and the output path for the transmit/receive shift register.

IO2FV Bit (Single-/Dual-SPI Mode QIO2 Output Fixed Value)

This bit sets the output value of the QIO2 pin in single-/dual-SPI modes. This bit is valid only in single-/dual-SPI modes, and is not affected by the MOIFE or MOIFV bit values.

IO3FV Bit (Single-/Dual-SPI Mode QIO3 Output Fixed Value)

This bit sets the output value of the QIO3 pin in single-/dual-SPI modes. This bit is valid only in single-/dual-SPI modes, and is not affected by the MOIFE or MOIFV bit values.

MOIFV Bit (Idle State Output Data Fixed Value)

If the MOIFE bit is 1, the pin output value during the QSSL negation period is determined according to this bit setting.

MOIFE Bit (Idle State Output Data Fixing Enable)

This bit fixes the pin output value in a QSSL negation period or the QSSL keeping period during a burst transfer. In single-SPI mode, this bit setting applies to QMO. In dual-SPI mode, this bit setting applies to QIO1 and QIO0. In quad-SPI mode, this bit setting applies to QIO3 through QIO0.

45.2.4 QSPI Status Register (SPSR)

Address(es): QSPI.SPSR 0008 9E03h

b7	b6	b5	b4	b3	b2	b1	b0
SPRFF	TREND	SPTEF	SPSSLF	—	—	—	—
0	1	1	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	SPSSLF	QSSL Negation Flag	0: QSSL negation is not detected. 1: QSSL negation is detected.	R/(W) *1
b5	SPTEF	Transmit Buffer Empty Flag	0: The number of transmit data units in the transmit buffer is greater than the transmit buffer data triggering number. 1: The number of transmit data units in the transmit buffer is equal to or less than the specified transmit buffer data triggering number.	R/(W) *1
b6	TREND	Transmit End Flag	0: Transmission is not completed. 1: Transmission is completed.	R
b7	SPRFF	Receive Buffer Full Flag	0: The number of receive data units in the receive buffer is less than the receive buffer data triggering number. 1: The number of receive data units in the receive buffer is equal to or greater than the receive buffer data triggering number.	R/(W) *1

Note 1. Only 0 can be written to clear the flag after reading 1.

The SPSR register holds flags indicating the operating state of this module.

SPSSLF Flag (QSSL Negation Flag)

This flag indicates the state of QSSL. This flag should be read after the receive buffer (SPRXB) is read.

[Setting conditions]

- QSSL output is changed from 0 to 1 when the SSLP.SSLP bi is 0.
- QSSL output is changed from 1 to 0 when the SSLP.SSLP bi is 1.

[Clearing conditions]

- After reading the SPSSLF flag = 1, 0 is written to it.
- Reset

SPTEF Flag (Transmit Buffer Empty Flag)

This flag indicates that the number of transmit data units in the transmit buffer is equal to or less than the transmit buffer data triggering number specified in the buffer control register. A flag-clearing operation takes precedence over a flag-setting operation.

[Setting conditions]

- When the number of transmit data units in the transmit buffer is equal to or less than the specified transmit buffer data triggering number.
- When transmit buffer reset is enabled.
- Reset

[Clearing conditions]

- When 0 is written to the SPTEF flag after reading SPTEF = 1.
- When data is written to the transmit buffer (SPTXB) using DTC or DMAC transfer (transfer of the last block in block transfer).

Note: Do not set the SPTEF flag to 0 during DTC or DMAC transfer.

TREND Flag (Transmit End Flag)

This flag is set to 1 when transmission is completed, and this flag is 0 when transmission is not completed. When the value of the SPCR.SPE bit is changed from 1 to 0, read this flag after waiting for the division ratio setting cycles plus four PCLK cycles.

For division ratio settings, see Table 45.3, SPBR Register and SPCMDn.BRDV[1:0] Bits Settings and Bit Rates (n = 0 to 3).

[Setting condition]

- When the number of data units stored in the transmit buffer is zero when a serial transmission is completed except when the SPDCR.TXDMY bit is set to 1.

[Clearing condition]

- When transmit data is transferred from the transmit buffer to the transmit shift register.

SPRFF Flag (Receive Buffer Full Flag)

This flag indicates that the number of receive data units in the receive buffer is equal to or greater than the receive buffer data triggering number specified in the buffer control register. A flag-clearing operation takes precedence over a flag-setting operation.

[Setting condition]

- The number of data units in the receive buffer is equal to or greater than the specified receive buffer data triggering number.

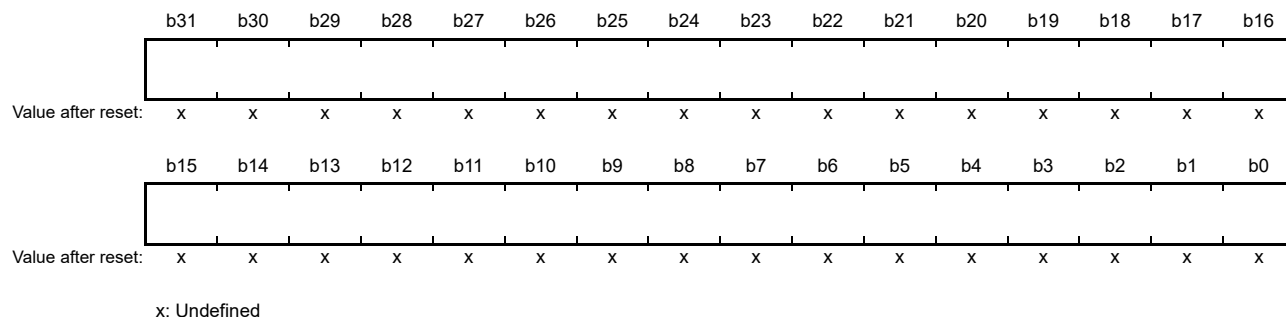
[Clearing conditions]

- When 0 is written to the SPRFF flag after reading SPRFF = 1.
- When data is read from the receive buffer (SPRXB) using DTC or DMAC transfer (transfer of the last block in block transfer). Do not set the SPRFF flag to 0 during DTC or DMAC transfer.

45.2.5 QSPI Data Register (SPDR)

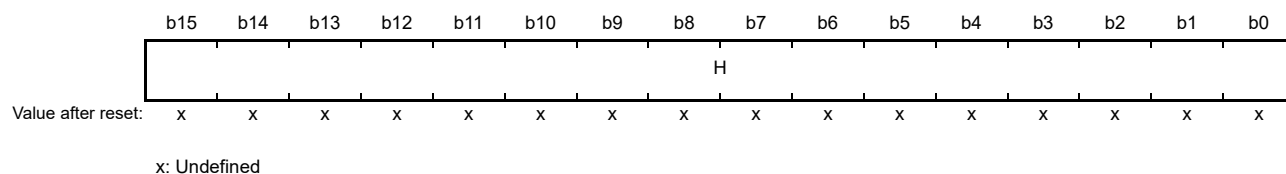
- When accessing in longword size

Address(es): QSPI.SPDR 0008 9E04h



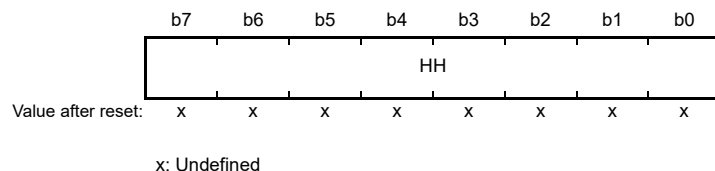
- When accessing in word size

Address(es): QSPI.SPDR.H 0008 9E04h



- When accessing in byte size

Address(es): QSPI.SPDR.HH 0008 9E04h



The SPDR register accesses transmit/receive data buffer.

The transmit buffer (SPTXB) and receive buffer (SPRXB) are independent and are mapped to the SPDR register.

When data is written to this register, the data will be written to the transmit buffer.

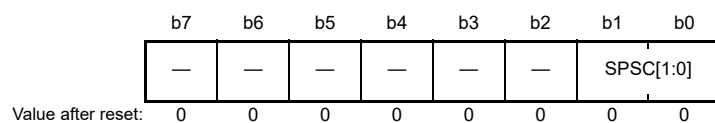
When data is read from this register, the data will be read from the receive buffer.

This register should be read or written to in byte, word, or longword units.

This register should be accessed in longword, word, or byte size.

45.2.6 QSPI Sequence Control Register (SPSCR)

Address(es): QSPI.SPSCR 0008 9E08h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	SPSC[1:0]	Sequence Control Specification	Number of SPCMDn register to be referenced (n = 0 to 3) b1 b0 0 0: 0 → 0 → ... 0 1: 0 → 1 → 0 → ... 1 0: 0 → 1 → 2 → 0 → ... 1 1: 0 → 1 → 2 → 3 → 0 → ...	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPSCR register sets the sequence controlled method. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

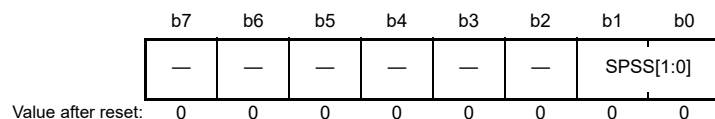
SPSC[1:0] Bits (Sequence Control Specification)

These bits specify sequential operations.

This module references the SPCMDn register (n = 0 to 3) in the order according to these bit settings.

45.2.7 QSPI Sequence Status Register (SPSSR)

Address(es): QSPI.SPSSR 0008 9E09h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	SPSS[1:0]	Sequence Status	b1 b0 0 0: SPCMD0 register 0 1: SPCMD1 register 1 0: SPCMD2 register 1 1: SPCMD3 register	R
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPSSR register indicates the sequence control status.

SPSS[1:0] Bits (Sequence Status)

During sequence control, these bits indicate the SPCMDn register (n = 0 to 3) that is currently referenced.

45.2.8 QSPI Bit Rate Register (SPBR)

Address(es): QSPI.SPBR 0008 9E0Ah

b7	b6	b5	b4	b3	b2	b1	b0
SPBR7	SPBR6	SPBR5	SPBR4	SPBR3	SPBR2	SPBR1	SPBR0
Value after reset:	1	1	1	1	1	1	1

The SPBR register sets the bit rate that is used as the base. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

The bit rate is determined by combinations of SPBR register settings and the bit settings in the SPCMDn.BRDV[1:0] bits (n = 0 to 3).

Setting this register to 0 is prohibited.

The equation for calculating the bit rate when SPBR is not 0 is given below. In the equation, m denotes an SPBR register setting (1 to 255), and N denotes bit settings (0 to 3) in the SPCMDn.BRDV[1:0] bits, and $2 \times m \times 2^N$ denotes a division ratio.

$$\text{Bit rate} = \frac{f(\text{PCLK})}{2 \times m \times 2^N}$$

Table 45.3 shows the relationship between SPBR register and SPCMDn.BRDV[1:0] bit settings and bit rates.

Table 45.3 SPBR Register and SPCMDn.BRDV[1:0] Bits Settings and Bit Rates (n = 0 to 3)

SPBR Register (m)	SPCMDn.BRDV[1:0] Bits (N)	Division Ratio	Bit Rate (PCLK = 60 MHz)
1	0	2	30.00 Mbps
2	0	4	15.00 Mbps
3	0	6	10.00 Mbps
4	0	8	7.50 Mbps
5	0	10	6.00 Mbps
6	0	12	5.00 Mbps
6	1	24	2.50 Mbps
6	2	48	1.25 Mbps
6	3	96	625 kbps
255	3	4080	14.70 kbps

Note: Set the SPBR register and the SPCMDn.BRDV[1:0] bits so that the division ratio of QSPCLK is 2 or more.

45.2.9 QSPI Data Control Register (SPDCR)

Address(es): QSPI.SPDCR 0008 9E0Bh

	b7	b6	b5	b4	b3	b2	b1	b0
	TXDMY	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	TXDMY	Dummy Data Transmission Enable	0: Disables dummy data transmission. 1: Enables dummy data transmission.	R/W

The SPDCR register enables or disables dummy data transmission.

TXDMY Bit (Dummy Data Transmission Enable)

This bit enables or disables dummy data transmission from the QMO pin in single-SPI mode when the transmit buffer is empty.

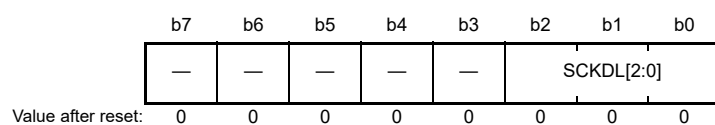
If the TXDMY bit is set to 1 when the transmit buffer is empty, setting the SPCR.SPE bit = 1 allows 0 to be output from the QMO pin as a dummy data. This enables starting of receive operation without writing transmit data to the transmit buffer.

If this bit is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

This bit is invalid in dual-/quad-SPI operations.

45.2.10 QSPI Clock Delay Register (SPCKD)

Address(es): QSPI.SPCKD 0008 9E0Ch



Bit	Symbol	Bit Name	Description	R/W																											
b2 to b0	SCKDL[2:0]	Clock Delay Setting	<table><tr><th>b2</th><th>b0</th><th></th></tr><tr><td>0</td><td>0</td><td>0: 1.5 QSPCLK</td></tr><tr><td>0</td><td>0</td><td>1: 2.5 QSPCLK</td></tr><tr><td>0</td><td>1</td><td>0: 3.5 QSPCLK</td></tr><tr><td>0</td><td>1</td><td>1: 4.5 QSPCLK</td></tr><tr><td>1</td><td>0</td><td>0: 5.5 QSPCLK</td></tr><tr><td>1</td><td>0</td><td>1: 6.5 QSPCLK</td></tr><tr><td>1</td><td>1</td><td>0: 7.5 QSPCLK</td></tr><tr><td>1</td><td>1</td><td>1: 8.5 QSPCLK</td></tr></table>	b2	b0		0	0	0: 1.5 QSPCLK	0	0	1: 2.5 QSPCLK	0	1	0: 3.5 QSPCLK	0	1	1: 4.5 QSPCLK	1	0	0: 5.5 QSPCLK	1	0	1: 6.5 QSPCLK	1	1	0: 7.5 QSPCLK	1	1	1: 8.5 QSPCLK	R/W
b2	b0																														
0	0	0: 1.5 QSPCLK																													
0	0	1: 2.5 QSPCLK																													
0	1	0: 3.5 QSPCLK																													
0	1	1: 4.5 QSPCLK																													
1	0	0: 5.5 QSPCLK																													
1	0	1: 6.5 QSPCLK																													
1	1	0: 7.5 QSPCLK																													
1	1	1: 8.5 QSPCLK																													
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																											

The SPCKD register sets a period (clock delay) from the beginning of QSSL signal assertion to QSPCLK oscillation when the SPCMDn.SCKDEN bit (n = 0 to 3) is 1.

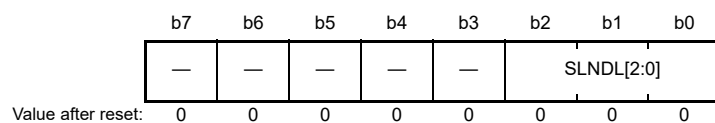
If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

SCKDL[2:0] Bits (Clock Delay Setting)

These bits set a period (clock delay) from the beginning of QSSL signal assertion to QSPCLK oscillation when the SPCMDn.SCKDEN bit is 1.

45.2.11 QSPI Slave Select Negation Delay Register (SSLND)

Address(es): QSPI.SSLND 0008 9E0Dh



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SLNDL[2:0]	QSSL Negation Delay Setting	b2 b0 0 0 0: 1 QSPCLK 0 0 1: 2 QSPCLK 0 1 0: 3 QSPCLK 0 1 1: 4 QSPCLK 1 0 0: 5 QSPCLK 1 0 1: 6 QSPCLK 1 1 0: 7 QSPCLK 1 1 1: 8 QSPCLK	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

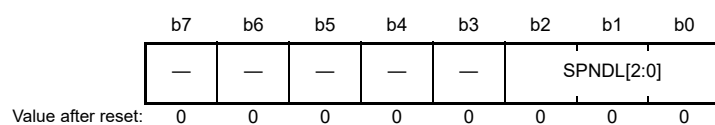
The SSLND register sets a period (QSSL negation delay) from a final QSPCLK edge to the negation of the QSSL signal during a serial transfer. when the SPCMDn.SLNDEN bit (n = 0 to 3) is 1. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

SLNDL[2:0] Bits (QSSL Negation Delay Setting)

These bits set a period (QSSL negation delay) from a final QSPCLK edge to the negation of the QSSL signal during a serial transfer when the SPCMDn.SLNDEN bit is 1.

45.2.12 QSPI Next-Access Delay Register (SPND)

Address(es): QSPI.SPND 0008 9E0Eh



Bit	Symbol	Bit Name	Description	R/W																											
b2 to b0	SPNDL[2:0]	Next-Access Delay Setting	<table><tr><th>b2</th><th>b0</th><th></th></tr><tr><td>0</td><td>0</td><td>0: 1 QSPCLK</td></tr><tr><td>0</td><td>0</td><td>1: 2 QSPCLK</td></tr><tr><td>0</td><td>1</td><td>0: 3 QSPCLK</td></tr><tr><td>0</td><td>1</td><td>1: 4 QSPCLK</td></tr><tr><td>1</td><td>0</td><td>0: 5 QSPCLK</td></tr><tr><td>1</td><td>0</td><td>1: 6 QSPCLK</td></tr><tr><td>1</td><td>1</td><td>0: 7 QSPCLK</td></tr><tr><td>1</td><td>1</td><td>1: 8 QSPCLK</td></tr></table>	b2	b0		0	0	0: 1 QSPCLK	0	0	1: 2 QSPCLK	0	1	0: 3 QSPCLK	0	1	1: 4 QSPCLK	1	0	0: 5 QSPCLK	1	0	1: 6 QSPCLK	1	1	0: 7 QSPCLK	1	1	1: 8 QSPCLK	R/W
b2	b0																														
0	0	0: 1 QSPCLK																													
0	0	1: 2 QSPCLK																													
0	1	0: 3 QSPCLK																													
0	1	1: 4 QSPCLK																													
1	0	0: 5 QSPCLK																													
1	0	1: 6 QSPCLK																													
1	1	0: 7 QSPCLK																													
1	1	1: 8 QSPCLK																													
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W																											

The SPND register sets a period (next-access delay) from termination of a serial transfer to the beginning of the next serial transfer when the SPCMDn.SPNDEN bit (n = 0 to 3) is 1. If this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

SPNDL[2:0] Bits (Next-Access Delay Setting)

These bits set a period (next-access delay) from termination of a serial transfer to the beginning of the next serial transfer when the SPCMDn.SPNDEN bit is 1.

45.2.13 QSPI Command Register n (SPCMDn) (n = 0 to 3)

Address(es): QSPI.SPCMD0 0008 9E10h, QSPI.SPCMD1 0008 9E12h, QSPI.SPCMD2 0008 9E14h, QSPI.SPCMD3 0008 9E16h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SCKDEN	SLNDEN	SPNDEN	LSBF	SPB[3:0]				SSLKP	SPIMOD[1:0]		SPRW	BRDV[1:0]		CPOL	CPHA
Value after reset:	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	CPHA	QSPCLK Phase Setting ^{*1}	0: Data latch on odd edge, data shift on even edge 1: Data shift on odd edge, data latch on even edge	R/W
b1	CPOL	QSPCLK Polarity Setting	0: Positive (QSPCLK = 0 when idle) 1: Negative (QSPCLK = 1 when idle)	R/W
b3, b2	BRDV[1:0]	Bit Rate Frequency Division Setting ^{*2}	b3 b2 0 0: Base bit rate 0 1: Base bit rate divided by 2 1 0: Base bit rate divided by 4 1 1: Base bit rate divided by 8	R/W
b4	SPRW	SPI Read/Write Access	0: Write operation (QIO1 and QIO0/QIO3 to QIO0: Output) 1: Read operation (QIO1 and QIO0/QIO3 to QIO0: Input)	R/W
b6, b5	SPIMOD[1:0]	SPI Operating Mode	b6 b5 0 0: Single-SPI 0 1: Dual-SPI 1 0: Quad-SPI 1 1: Setting prohibited	R/W
b7	SSLKP	QSSL Signal Level Keeping	0: Negates all QSSL signals upon completion of transfer. 1: Keeps the QSSL signal level from the end of the transfer to the beginning of the next access.	R/W
b11 to b8	SPB[3:0]	Transfer Data Length Setting	b11 b8 0 0 0 0: 8 bits (1 byte) 0 0 0 1: 16 bits (2 bytes) 0 0 1 0: 32 bits (4 bytes) Settings other than above are prohibited.	R/W
b12	LSBF	LSB First	0: MSB first 1: LSB first	R/W
b13	SPNDEN	Next-Access Delay Enable	0: A next-access delay of 0 QSPCLK cycle 1: A next-access delay equal to SPND register settings.	R/W
b14	SLNDEN	QSSL Negation Delay Setting Enable	0: A QSSL negation delay of 0 QSPCLK cycle 1: A QSSL negation delay equal to SSLND register settings.	R/W
b15	SCKDEN	Clock Delay Setting Enable	0: A clock delay of 0 QSPCLK cycle 1: A clock delay equal to SPCKD register settings.	R/W

Note: When setting any or all of the clock delay period, QSSL negation delay period, and next-access delay period to 0, set the SSLKP bit to 1 to select the continuous access in which QSSL is not negated. Otherwise, operation cannot be guaranteed. For the method of setting the various delay periods for the continuous access in which QSSL is not negated, see below.

Note: For the continuous access in which QSSL is not negated, QSPCLK clock stopping is followed by the QSSL negation delay period, next-access delay period, and next command clock delay period, in this order. There are restrictions on setting the SCKDEN, SLNDEN, and SPNDEN bits at this time. For details, see the SCKDEN bit description.

Note 1. The first QSPCLK edge is treated as the first edge.

Note 2. Set the SPBR register and the SPCMDn.BRDV[1:0] bits so that the division ratio of QSPCLK is 2 or more.

The SPCMDn register (n = 0 to 3) is used to set a transfer format. This module sequentially references the SPCMDn register according to the settings in the SPSCR register, and executes the serial transfer that is set in the referenced the SPCMDn register.

If the SPCMDn register currently being referred to is modified while the SPE bit is 1, subsequent operation cannot be guaranteed. The SPCMDn register currently being referred to can be checked by reading the SPSSR register.

CPHA Bit (QSPCLK Phase Setting)

This bit sets a QSPCLK edge for latching and shifting data to be transferred. The phase should be the same as that of the device to be communicated with.

This MCU supports four SPI modes listed in the Table 45.4 in combination of the CPHA bit with the CPOL bit.

CPOL Bit (QSPCLK Polarity Setting)

This bit sets a QSPCLK polarity. The polarity should be same as that of the device to be communicated with.

This MCU supports four SPI modes listed in the Table 45.4 in combination of the CPOL bit with the CPHA bit.

Table 45.4 SPI Modes Definitions

SPI Mode	CPOL	CPHA
Mode 0	0	0
Mode 1	0	1
Mode 2	1	0
Mode 3	1	1

BRDV[1:0] Bits (Bit Rate Frequency Division Setting)

The settings of these bits and of the SPBR register together determine the bit rate. The base bit rate depends on the setting of the SPBR register. The setting of these bits selects division of the base bit rate by 1, 2, 4 or 8.

Note: In transmission, set the SPBR register and these bits so that the division ratio of QSPCLK is 2 or more.

Note: In communication and burst transfer using sequence control, the set value of the BRDV[1:0] bits cannot be changed for each of the commands that configure the sequence.

SPRW Bit (SPI Read/Write Access)

This bit sets an access direction in dual-/quad-SPI modes.

This bit is invalid in single-SPI mode.

SPIMOD[1:0] Bits (SPI Operating Mode)

These bits select the operating mode from single-, dual-, or quad-SPI.

SSLKP Bit (QSSL Signal Level Keeping)

This bit specifies whether the QSSL signal level for the current command is to be kept or not from the end of the transfer for the current command to the beginning of the transfer for the next command. Setting this bit to 1 enables a transition to the next access while the QSSL signal is kept asserted.

SPB[3:0] Bits (Transfer Data Length Setting)

These bits set the basic transfer data length for serial transfer. For LSB first transfer, the transfer data is reversed within the data width specified with these bits. The actual amount of data to be transferred is determined by multiplying the value set with these bits by the value set with the SPBMULn register (n = 0 to 3).

LSBF Bit (LSB First)

This bit sets the data format to MSB first or LSB first.

SPNDEN Bit (Next-Access Delay Enable)

This bit sets the period (next-access delay) from termination of a serial transfer to the beginning of the next serial transfer. If this bit is 0, this module sets the next-access delay to 0 QSPCLK cycle. If this bit is 1, this module starts next

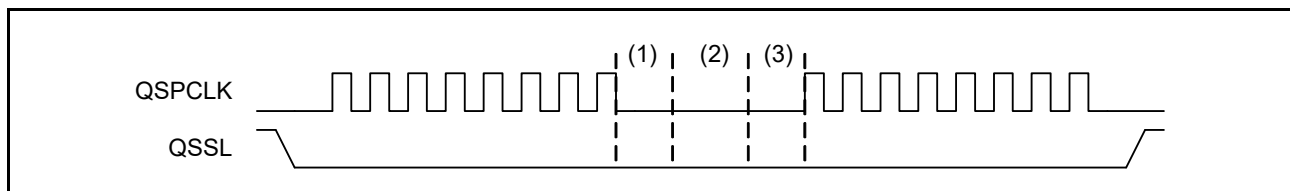
serial transfer in compliance with the SPND register settings. For the continuous access in which QSSL is kept asserted over the multiple commands, this bit can be set to 0 only when the pertinent command is not the last one. Otherwise, this bit should be set to 1.

SLNDEN Bit (QSSL Negation Delay Setting Enable)

This bit sets a period (QSSL negation delay) from QSPCLK oscillation stoppage to QSSL signal negation. If this bit is 0, this module sets the QSSL negation delay to 0 QSPCLK cycle. If this bit is 1, this module negates the QSSL signal in compliance with the SSLND register settings. For the continuous access in which QSSL is kept asserted over the multiplier commands, this bit can be set to 0 only when the pertinent command is not the last one. Otherwise, this bit should be set to 1.

SCKDEN Bit (Clock Delay Setting Enable)

This bit sets a period (clock delay) from the beginning of QSSL signal assertion to QSPCLK oscillation. If this bit is 0, this module sets the clock delay to 0 QSPCLK cycle. If this bit is 1, this module starts QSPCLK oscillation in compliance with the SPCKD register settings. For the continuous access in which QSSL is kept asserted over the multiple commands, this bit can be set to 0 only when the pertinent command is the second or subsequent one. Otherwise, this bit should be set to 1.

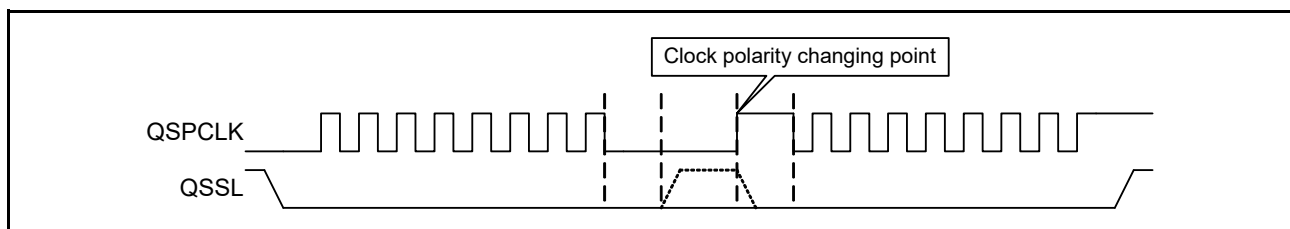


In the above figure, (1), (2), and (3) refer to the QSSL negation delay period, next-access delay period, and next command clock delay period, respectively.

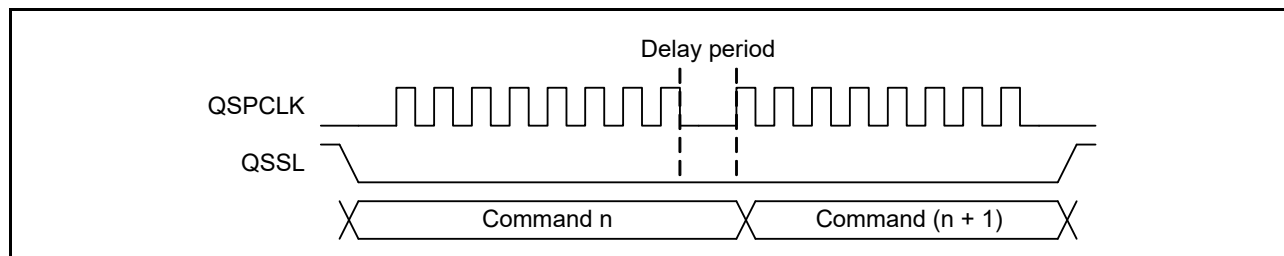
Allowed setting is $\{(1), (2), (3)\} = \{1, 1, 1\}, \{1, 1, 0\}, \{1, 0, 0\}, \{0, 0, 0\}$. If a combination other than these is set, operation cannot be guaranteed.

Note: When changing the BRDV[1:0] bits or CPOL bit for each command for the continuous access in which the QSSL level is held, insert the QSSL negation delay period, next-access delay period, and clock delay period between commands. Otherwise, operation cannot be guaranteed.

Note: A clock polarity changing point may be detected as a clock edge if the CPOL bit is changed with the QSSL level held.

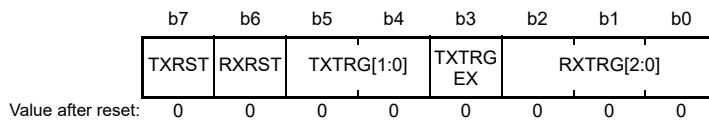


Note: When changing the SPIMOD[1:0] bits or CPHA bit for each command for the continuous access in which the QSSL level is held, insert one cycle or more between commands. Otherwise, operation cannot be guaranteed. (This also applies to write-to-read or read-to-write switching in dual-/quad-SPI modes.)
In the figure below, the data line is driven during the command n period if command n is for dual-/quad-SPI write access.



45.2.14 QSPI Buffer Control Register (SPBFCR)

Address(es): QSPI.SPBFCR 0008 9E18h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	RXTRG[2:0]	Receive Buffer Data Triggering Number	b2 b0 0 0 0: 1 byte (31 bytes available) 0 0 1: 2 bytes (30 bytes available) 0 1 0: 4 bytes (28 bytes available) 0 1 1: 5 bytes (27 bytes available) 1 0 0: 8 bytes (24 bytes available) 1 0 1: 16 bytes (16 bytes available) 1 1 0: 24 bytes (8 bytes available) 1 1 1: 32 bytes (0 byte available)	R/W
b3	TXTRGEX	Transmit Buffer Data Triggering Number	b5 b3 0 0 0: 31 bytes (1 byte available) 0 1 0: 30 bytes (2 bytes available) 1 0 0: 28 bytes (4 bytes available) 1 1 0: 0 byte (32 bytes available)	R/W
b5, b4	TXTRG[1:0]		0 0 1: 24 bytes (8 bytes available) 0 1 1: 16 bytes (16 bytes available) 1 0 1: 8 bytes (24 bytes available) 1 1 1: 4 bytes (28 bytes available)	
b6	RXRST	Receive Buffer Data Reset	0: Allows the receive buffer normal operation. 1: Resets the receive buffer.	R/W
b7	TXRST	Transmit Buffer Data Reset	0: Allows the transmit buffer normal operation. 1: Resets the transmit buffer.	R/W

The SPBFCR register resets the number of data units in the transmit buffer (SPTXB) or receive buffer (SPRXB) and sets the number of triggering data units. If this register is modified, perform a dummy read after data is written, and then execute the subsequent instruction. In addition, if this register is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed.

RXTRG[2:0] Bits (Receive Buffer Data Triggering Number)

These bits specify the timing at which the receive buffer full state is determined, that is when the SPSR.SPRFF flag in the status register is set. When the number of bytes of data in the receive buffer (SPRXB) is equal to or greater than the specified triggering number, the SPSR.SPRFF flag is set to 1. If these bits are modified, perform a dummy read after data is written, and then execute the subsequent instruction. These bits should be modified when the SPCR.SPE bit is 0.

TXTRG[1:0] Bits and TXTRGEX Bit (Transmit Buffer Data Triggering Number)

These bits specify the timing at which the transmit buffer empty state is determined, that is when the SPSR.SPRFF flag in the status register is set. When the number of bytes of data in the transmit buffer (SPTXB) is equal to or less than the specified triggering number, the SPSR.SPTEF flag is set to 1. If these bits are modified, perform a dummy read after data is written, and then execute the subsequent instruction. These bits should be modified when the SPCR.SPE bit is 0.

RXRST Bit (Receive Buffer Data Reset)

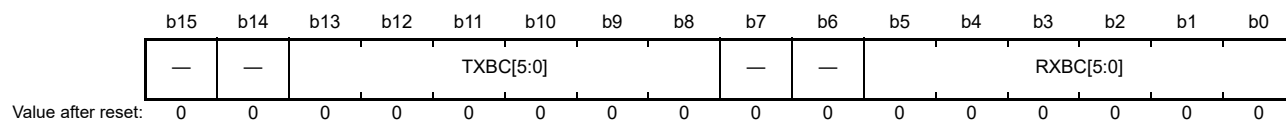
This bit invalidates receive data in the receive buffer and resets the receive buffer to an empty state. If this bit is modified, perform a dummy read after data is written, and then execute the subsequent instruction. This bit should be modified when the SPCR.SPE bit is 0.

TXRST Bit (Transmit Buffer Data Reset)

This bit invalidates transmit data in the transmit buffer and resets the transmit buffer to an empty state. If this bit is modified, perform a dummy read after data is written, and then execute the subsequent instruction. This bit should be modified when the SPCR.SPE bit is 0.

45.2.15 QSPI Buffer Data Count Set Register (SPBDCR)

Address(es): QSPI.SPBDCR 0008 9E1Ah



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	RXBC[5:0]	Receive Data Byte Counter	000000b indicates that SPRXB is empty. 100000b indicates that SPRXB is full.	R
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13 to b8	TXBC[5:0]	Transmit Data Byte Counter	000000b indicates that SPTXB is empty. 100000b indicates that SPTXB is full.	R
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPBDCR register indicates the number of data units stored in the transmit buffer (SPTXB) and receive buffer (SPRXB). The upper eight bits indicate the number of transmit data units in the transmit buffer and the lower 8 bits indicate the number of receive data units in the receive buffer.

RXBC[5:0] Bits (Receive Data Byte Counter)

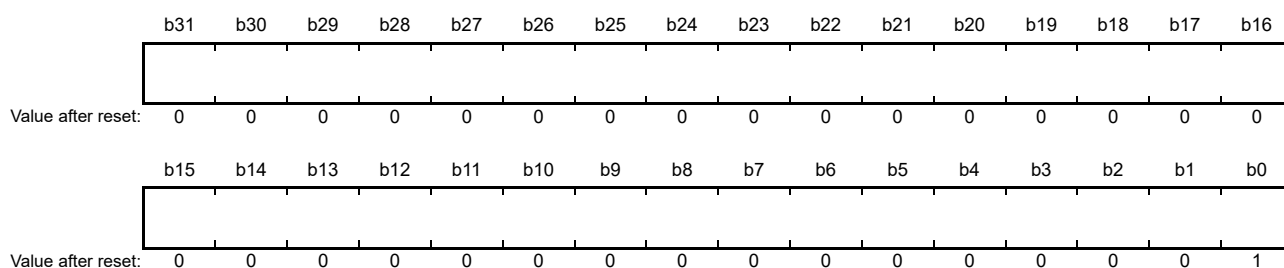
These bits indicate the number of receive data bytes in the receive data buffer (SPRXB).

TXBC[5:0] Bits (Transmit Data Byte Counter)

These bits indicate the number of transmit data bytes in the transmit data buffer (SPTXB).

45.2.16 QSPI Transfer Data Length Multiplier Setting Register n (SPBMULn) (n = 0 to 3)

Address(es): QSPI.SPB MUL0 0008 9E1Ch, QSPI.SPB MUL1 0008 9E20h, QSPI.SPB MUL2 0008 9E24h, QSPI.SPB MUL3 0008 9E28h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	Transfer Data Length Multiplier Setting	These bits set the multiplier for the transfer data length.	R/W

The SPBMULn register (n = 0 to 3) sets the number of times to transfer the specific length of data defined by the SPCMDn.SPB[3:0] bits. The SPBMUL0 to SPBMUL3 registers correspond to the SPCMD0 to SPCMD3 registers, respectively.

The actual amount of data to be transferred is determined by the SPCMDn.SPB[3:0] bits × the SPBMULn register. Setting this register to 0000 0000h allows the defined size of data to be transferred 4,294,967,296 times.

If the SPBMULn register corresponding to the SPCMDn register currently being referred to is modified while the SPCR.SPE bit is 1, subsequent operation cannot be guaranteed. The SPCMDn register currently being referred to can be checked by reading the SPSSR register.

45.3 Operation

In this section, the serial transfer period means a period from the beginning of driving valid data to the fetching of the final valid data, and the idle period means the QSSL negation period.

45.3.1 Overview of Operations

This module is capable of serial transfers in single-/dual-/quad-SPI modes. Table 45.5 shows the features of each SPI mode.

Table 45.5 Features of Each SPI Mode

	Single-SPI	Dual-SPI	Quad-SPI
Number of data lines	One input line and one output line	Two I/O lines	Four I/O lines
Data line direction	Single-directional	Bidirectional	Bidirectional
Simultaneous transmission/reception	Supported	Not supported	Not supported

Table 45.6 shows the overview of operation.

Table 45.6 Overview of Operation

Items	Specification
QSPCLK signal	Output
QMO signal (single-SPI)	Output
QMI signal (single-SPI)	Input
QIO1 and QIO0 (dual-SPI)/QIO3 to QIO0 (quad-SPI)	Input/output
QSSL signal	Output
Switching QSSL polarity	Supported
Transfer bit rate	Up to PCLK/2
Clock source	On-chip baud rate generator
Clock polarity	Positive/negative
Clock phase	Data latch at odd edge, data shift at even edge/ data shift at odd edge, data latch at even edge
Transfer bit order	MSB first/LSB first
Transfer data length	$(8/16/32) \times (1 \text{ to } 4,294,967,296) \text{ bits}$
Burst transfer	Supported
QSPCLK delay control	Supported
QSSL negation delay control	Supported
Next-access delay control	Supported
Transfer start method	The specified length of data is in the transmit buffer when the SPCR.SPE bit = 1 There is a space for the length of data in the receive buffer when the SPCR.SPE bit = 1*1
Sequence control	Supported
Transmit buffer empty detection	Supported
Receive buffer full detection	Supported
QSSL negation detection	Supported

Note 1. In single-SPI operation, a transfer is started when the specified length of data is in the transmit buffer and there is a space for the specified length of data in the receive buffer. In dual-/quad-SPI operation, a transfer is started when the specified length of data is in the transmit buffer during transmission, and there is a space for the specified length of data in the receive buffer during reception.

45.3.2 Pin Control

This module automatically switches the pin states according to the status after write/read transfer in single-/dual-/quad-SPI mode. The status of the data pins (QMO/QMI/QIO[3:0]) in the idle state depends on the SPPCR.MOIFE, MOIFV, IO3FV, and IO2FV bit settings. Table 45.7 shows the pin states in single-SPI mode. Table 45.8 shows the pin states in dual-/quad-SPI modes.

Table 45.7 Pin States in Single-SPI Mode

Items	Single-SPI
QSSL	Output
QSPCLK	Output
QMO	Output
QMI	Input
QMO in the idle state	SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value
QMI in the idle state	—
QIO2	SPPCR.IO2FV bit setting value output or not used
QIO3	SPPCR.IO3FV bit setting value output or not used

Table 45.8 Pin States in Dual-/Quad-SPI Mode

Items	Dual-SPI	Quad-SPI
QSSL	Output	Output
QSPCLK	Output	Output
QIO0	Input/output	Input/output
QIO1	Input/output	Input/output
QIO2	SPPCR.IO2FV bit setting value output or not used	Input/output
QIO3	SPPCR.IO3FV bit setting value output or not used	Input/output
QIO0 in the idle state	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z
QIO1 in the idle state	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z
QIO2 in the idle state	SPPCR.IO2FV bit setting value output or not used	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z
QIO3 in the idle state	SPPCR.IO3FV bit setting value output or not used	After writing: SPPCR.MOIFE bit = 0: Final output value SPPCR.MOIFE bit = 1: SPPCR.MOIFV bit setting value After reading: Hi-Z

45.3.3 Transfer Format

The SPI has four modes determined by setting the SPCMDn.CPOL and SPCMDn.CPHA bits ($n = 0$ to 3). Figure 45.2 shows the data latch/shift timing based on each mode in an 8-bit MSB first transfer. In Figure 45.2, L indicates the latch timing and S indicates the shift timing. DATA corresponds to QMI/QMO in single-SPI mode; QIO1 and QIO0 in dual-SPI mode; or QIO3 to QIO0 in quad-SPI mode. t_{ckd} indicates the clock delay period when the SPCMDn.SCKDEN bit is set to 1. Similarly, t_{slnd} indicates the QSSL negation delay period when the SPCMDn.SLNDEN bit is set to 1, and t_{spnd} indicates the next-access delay period when the SPCMDn.SPNDEN bit is set to 1.

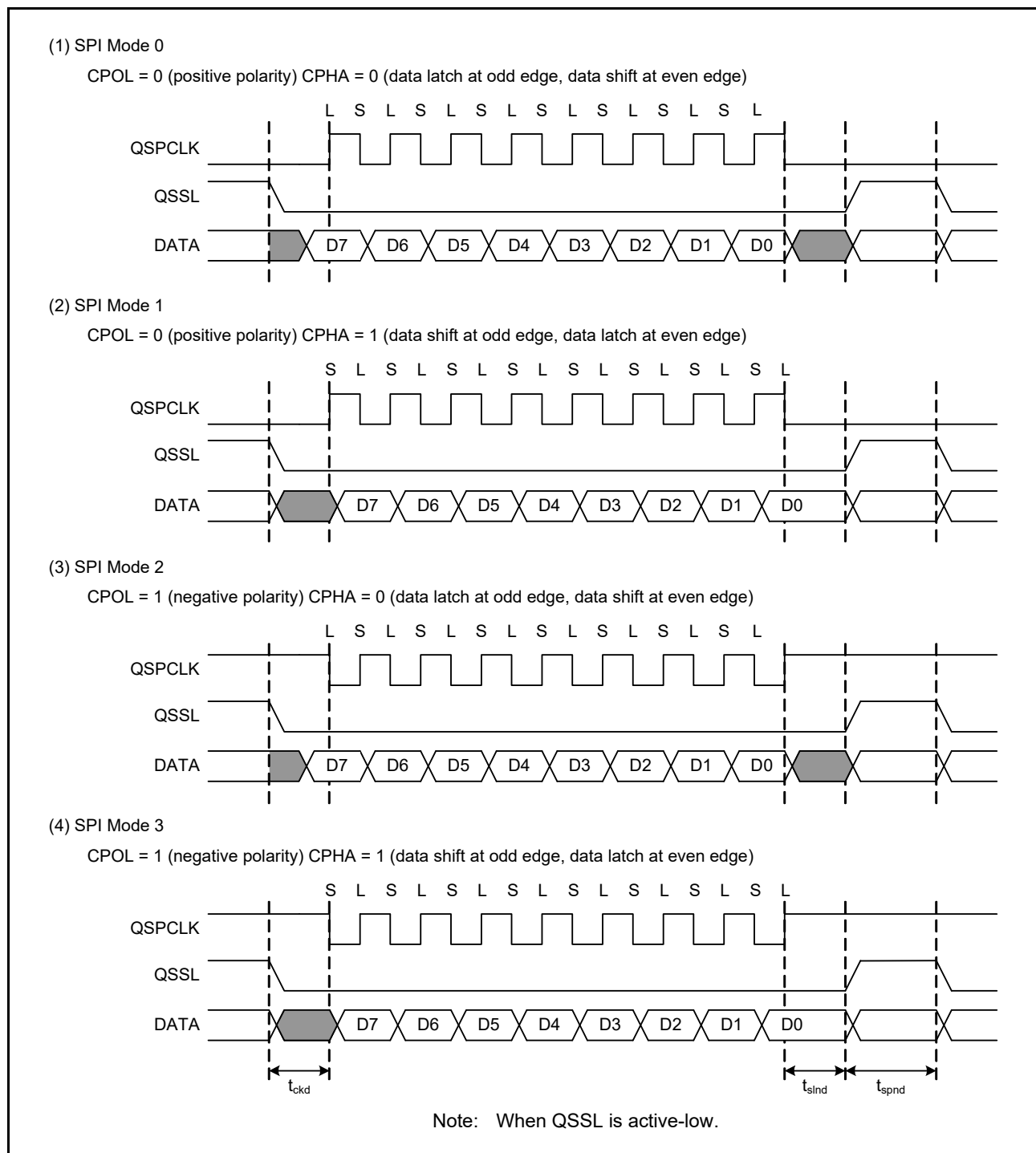


Figure 45.2 Clock Setting and Transfer Timing

Note that when the base bit rate is used, transmission and reception when the SPCMDn.CPHA bit = 0 is not available (n = 0 to 3).

The following describes 8-bit MSB first transfer in single-/dual-/quad-SPI modes when the SPCMDn.CPOL bit = 0 and the SPCMDn.CPHA bit = 0.

(1) Single-SPI Mode

Figure 45.3 shows the transfer format in single-SPI mode. This mode provides transmission and reception simultaneously. Since one data line is used for serial communication both in transmission and reception, the communication speed is 1 bit per QSPCLK clock cycle. Transfer data is specified using the SPCMDn register (n = 0 to 3). For details of transfer data, see section 45.3.4, Transfer Data.

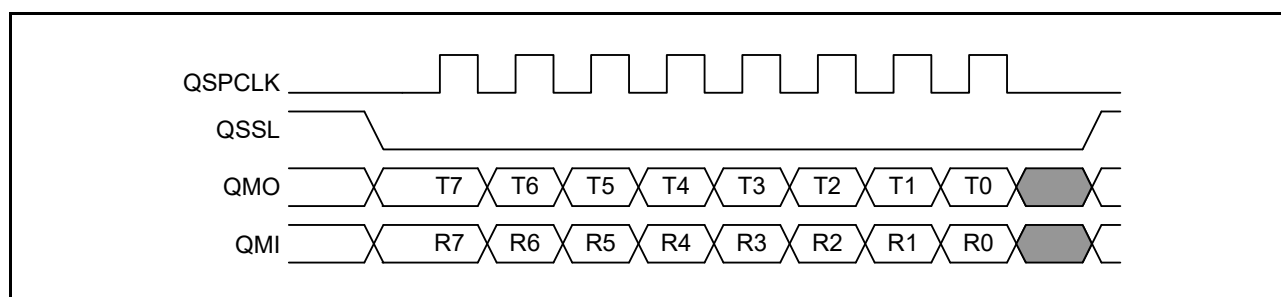


Figure 45.3 Transfer Format in Single-SPI Format

(2) Dual-SPI Mode

Figure 45.4 shows the transfer format in dual-SPI mode. This mode only provides operation of a single direction, that is, either transmission or reception. Transmission or reception can be set using the SPCMDn.SPRW bit. Transmission is carried out by write operation and reception by read operation. The I/O directions of QIO1 and QIO0 are switched accordingly. Since two data lines are used for serial communication both in transmission and reception, the communication speed is 2 bits per QSPCLK clock cycle. The start bit of the transfer data is output from QIO1. Transfer data is specified using the SPCMDn register. For details of transfer data, see section 45.3.4, Transfer Data.

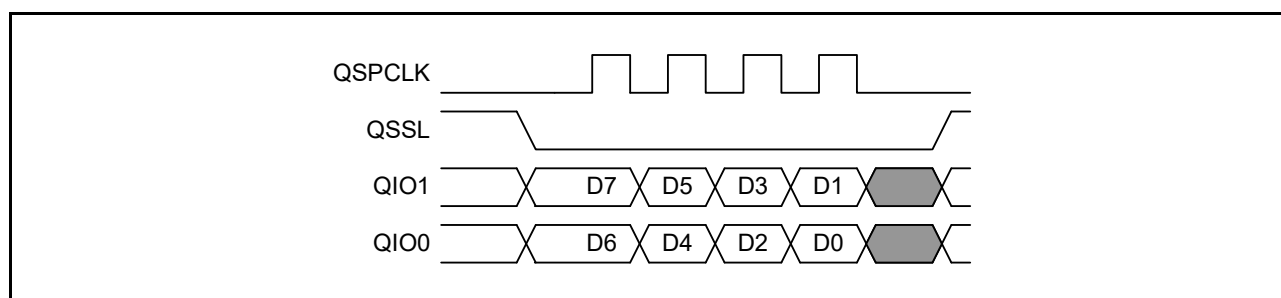


Figure 45.4 Transfer Format in Dual-SPI Format

(3) Quad-SPI Mode

Figure 45.5 shows the transfer format in quad-SPI mode. This mode provides operation of a single direction, that is, either transmission or reception. Transmission or reception can be set using the SPCMDn.SPRW bit ($n = 0$ to 3).

Transmission and reception are carried out by writing and reading, respectively. The I/O directions of QIO3 to QIO0 are switched accordingly. Since four data lines are used for serial communication both in transmission and reception, the communication speed is 4 bits per QSPCLK clock cycle. The start bit of the transfer data is output from QIO3. Transfer data is specified using the SPCMDn register. For details of transfer data, see section 45.3.4, Transfer Data.

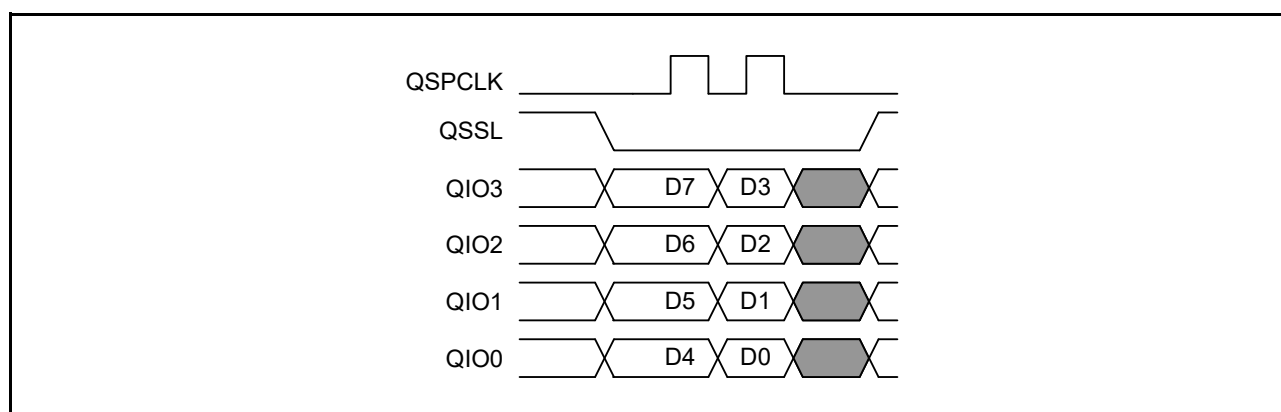


Figure 45.5 Transfer Format in Quad-SPI Format

45.3.4 Transfer Data

The data format is determined by the SPCMDn.SPB[3:0] bits and LSBF bit ($n = 0$ to 3) and the SPBMULn register. This module treats the specified size of data beginning at the MSB of the transmit shift register as transmit data, and the specified length of data beginning at the LSB of the receive shift register as receive data, regardless of whether the actual arrangement is MSB or LSB first. The following sections describe MSB first and LSB first transfers in 32-bit, 16-bit, and 8-bit data units.

(1) MSB First Transfer (32-Bit Data)

Figure 45.6 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs a 32-bit MSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 32-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module copies the data in the transmit buffer to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 32 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 32 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 32 bits or more, this module copies the 32-bit data beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 32 bits or more, data reception is not carried out. In order to start reception, the specified length of data should be read from the receive buffer to secure the space for 32 bits or more in the receive buffer.

In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

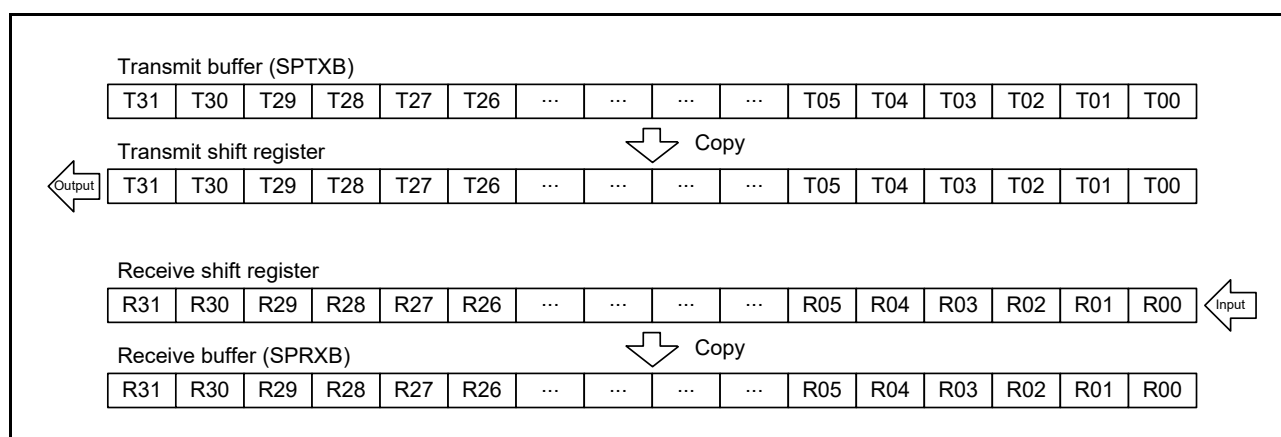


Figure 45.6 MSB First Transfer (32-Bit Data)

(2) MSB First Transfer (16-Bit Data)

Figure 45.7 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs a 16-bit MSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 16-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module copies the data with MSB-aligned in the transmit buffer to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 16 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 16 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 16 bits or more, this module copies the 16-bit data beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 16 bits or more, data reception is not carried out. In order to start reception, the specified length of data should be read from the receive buffer to secure the space for 16 bits or more in the receive buffer.

In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

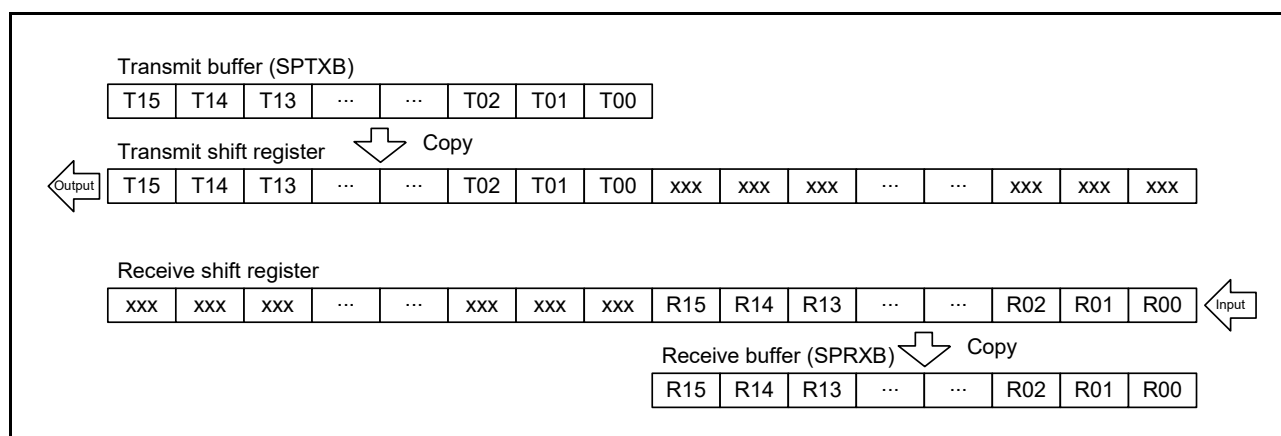


Figure 45.7 MSB First Transfer (16-Bit Data)

(3) MSB First Transfer (8-Bit Data)

Figure 45.8 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs an 8-bit MSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 8-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module copies the data with MSB-aligned in the transmit buffer to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 8 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 8 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 8 bits or more, this module copies the 8-bit data beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 8 bits or more, data reception is not carried out. In order to start reception, data for the specified length of data should be read from the receive buffer to secure the space for 8 bits or more in the receive buffer.

In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

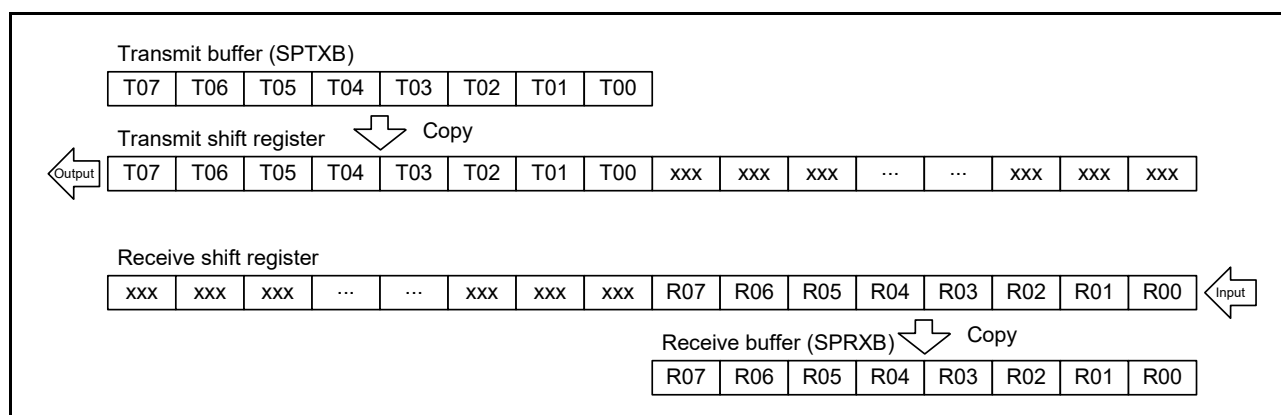


Figure 45.8 MSB First Transfer (8-Bit Data)

(4) LSB First Transfer (32-Bit Data)

Figure 45.9 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs a 32-bit LSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 32-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module reverses the bit order in the 32-bit transmit data, copies it to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 32 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 32 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 32 bits or more, this module reverses the order of the bits of the 32-bit data, copies it beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 32 bits or more, data reception is not carried out. In order to start reception, the specified length of data should be read from the receive buffer to secure the space for 32 bits or more in the receive buffer.

In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

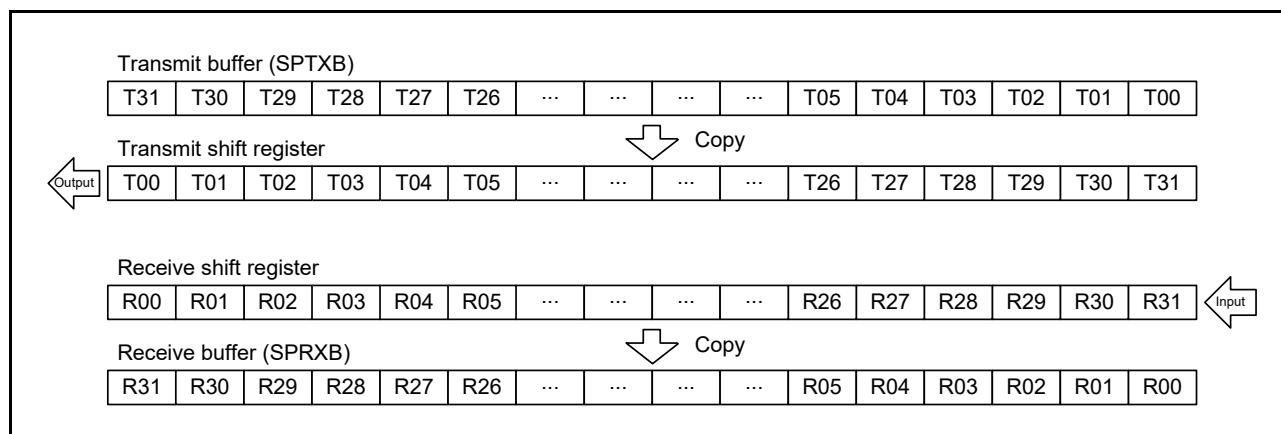


Figure 45.9 **LSB First Transfer (32-Bit Data)**

(5) LSB First Transfer (16-Bit Data)

Figure 45.10 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs a 16-bit LSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 16-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module reverses the bit order in the 16-bit transmit data, copies it with MSB-aligned to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 16 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 16 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 16 bits or more, this module reverses the bit order in the 16-bit data, copies it beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 16 bits or more, data reception is not carried out. In order to start reception, the specified length of data should be read from the receive buffer to secure the space for 16 bits or more in the receive buffer.

In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

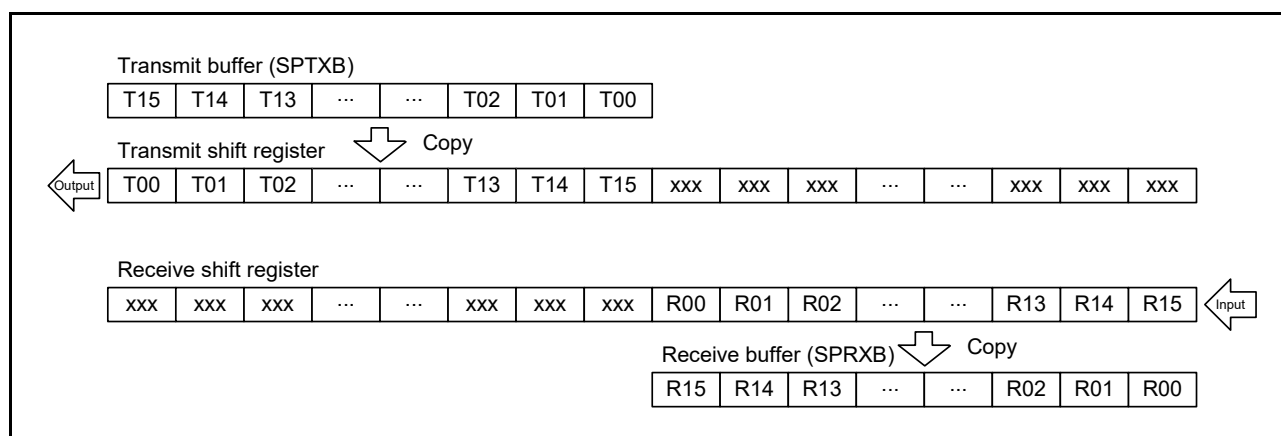


Figure 45.10 **LSB First Transfer (16-Bit Data)**

(6) LSB First Transfer (8-Bit Data)

Figure 45.11 shows the operation of the transmit buffer and transmit shift register, and the receive shift register and receive buffer when this module performs an 8-bit LSB-first data transfer.

For data transmission, the CPU, DTC, or DMAC writes 8-bit transmit data to the transmit buffer (SPTXB). If the transmit shift register is empty, this module reverses the bit order in the 8-bit transmit data, copies it with MSB-aligned to the transmit shift register, and fills the transmit shift register. When data transmission is started, this module outputs data beginning at the MSB (bit 31) of the transmit shift register, and when the QSPCLK clock cycle required for the serial transfer of 8 bits has passed, the transmit shift register becomes empty.

For data reception, data received from the data pin is stored in the receive shift register beginning at the LSB (bit 0). When the QSPCLK clock cycle required for the serial transfer of 8 bits has passed, the receive shift register becomes full. If the receive buffer (SPRXB) has a space for 8 bits or more, this module reverses the bit order in the 8-bit data, copies it beginning at the LSB from the receive shift register to the receive buffer, and empties the receive shift register. If the receive buffer does not have a space for 8 bits or more, data reception is not carried out. In order to start reception, the specified length of data should be read from the receive buffer to secure the space for 8 bits or more in the receive buffer. In actual transfer, this operation is repeated for the number of times defined by the SPBMULn register ($n = 0$ to 3).

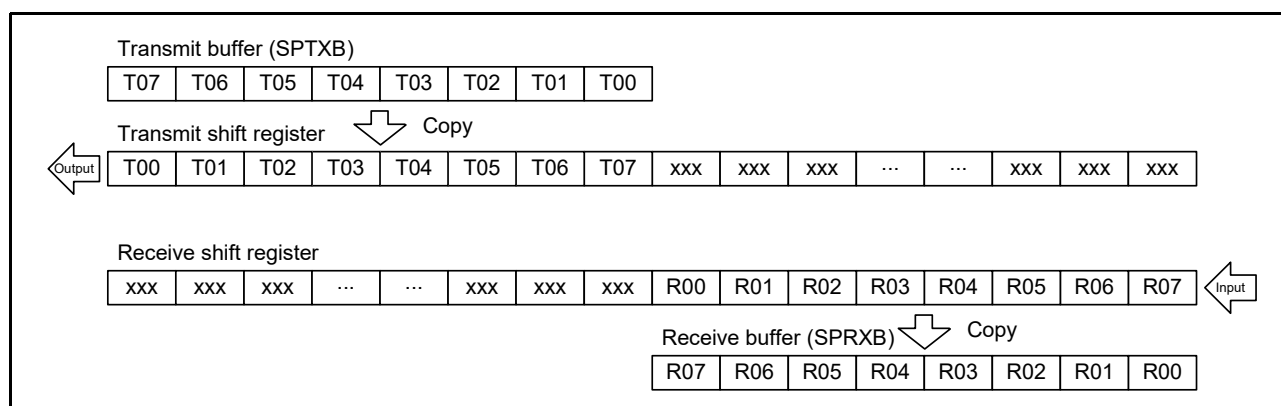


Figure 45.11 LSB First Transfer (8-Bit Data)

45.3.5 Non-Normal Transfer Operations

In the normal serial transfer, the data written from the SPDR register to the transmit buffer is serially transmitted, and the serially received data can be read from the receive buffer of the SPDR register. If access is made to the SPDR register, depending on the status of the transmit buffer/receive buffer, in some cases non-normal transfers can be executed.

Table 45.9 shows the relationship between non-normal transfer operations.

Table 45.9 Relationship between Non-Normal Transfer Operations

	Occurrence Condition	Operation
A	The SPDR register is written when the transmit buffer is full.	Missing write data.
B	The SPDR register is read when the receive buffer is empty.	The output data is undefined.

On operation A shown in Table 45.9, whether the SPDR register can be written to or not can be checked using the SPBDCR.TXBC[5:0] bits.

Similarly, on operation B shown in Table 45.9, whether the valid data is stored in the receive buffer or not can be checked by reading the SPBDCR.RXBC[5:0] bits.

45.3.6 Initialization

If 0 is written to the SPCR.SPE bit, this module disables the module function, and initializes a part of the module function. When a power-on reset is generated, this module initializes all of the module function.

When the SPCR.SPE bit is set to 0, this module performs the following initialization:

- Suspending any serial transfer that is being executed
- Initializing the transmit shift register and the receive shift register
- Initializing the internal state machine
- Initializing the sequence
- Initializing the SPSR.TREND flag

Initialization by the setting of the SPCR.SPE bit to 0 does not initialize the control bits of this module and the transmit/receive buffer. For this reason, this module can be started in the same transfer state as prior to the initialization if the SPCR.SPE bit is re-set to 1. However, setting the SPCR.SPE bit to 0 initializes the transmit shift register and the receive shift register, thus the data that is being transferred will be discarded.

45.3.7 SPI Operation

The operating modes of this module are listed below.

- Single-SPI mode
- Dual-SPI mode/quad-SPI mode

The operation in each mode is described below.

(1) Single-SPI Mode

(a) Starting Serial Transfer

The serial transfer start conditions are: there is the specified length of data in the transmit buffer; and there is space for the specified length of data in the receive buffer.

(b) Terminating Serial Transfer

Irrespective of the clock setting, this module terminates the serial transfer after transmitting an QSPCLK edge corresponding to the final sampling timing. After the serial transfer is completed, receive data is copied from the receive shift register to the receive buffer. If there is not enough space for the specified length of data in the receive buffer after receive data is copied from the receive shift register to the receive buffer, another serial transfer will not be performed.

(c) Sequence Control

In single-SPI mode, according to the sequence length that is assigned to the SPSCR register, this module makes up a sequence comprised of a part or all of the SPCMDn register (n = 0 to 3) and the SPBMULn register. This module contains a pointer to the SPCMDn register that makes up the sequence. The value of this pointer can be checked by reading the SPSSR register.

When the SPCR.SPE bit is set to 1 and the function of this module is enabled, this module loads the pointer to the commands in the SPCMD0 register, and incorporates the SPCMD0 and SPBMUL0 register settings into the transfer format at the beginning of serial transfer. This module increments the pointer each time the next-access delay period for a data transfer that corresponds to the referenced the SPCMDn register ends. Upon completion of the serial transfer that corresponds to the final command comprising the sequence, this module sets the pointer in the SPCMD0 register, and in this manner the sequence is executed repeatedly.

The following items are set in the SPCMDn register: basic transfer data length, MSB or LSB first, clock settings, some of the bit rate settings, SPI transfer mode and transfer direction (only in dual-/quad-SPI modes), whether QSSL level is held, a clock delay period, an QSSL negation delay period, and a next-access delay period. The total amount of data to be transferred is determined by multiplying the basic length of data to be transferred by the value set with the SPBMULn register.

Figure 45.12 shows an operation example when the SPSCR register is set to 02h, and the sequence is configured based on SPCMD0 to SPCMD2 register settings. In Figure 45.12, shaded areas of QMO/QMI indicate invalid data. Periods (1) to (3) in the figure indicate the followings.

- (1) Clock delay period (SPCKD) setting value = 000b (1.5 QSPCLK cycles)
- (2) QSSL negation delay period (SSLND) setting value = 000b (1 QSPCLK cycle)
- (3) Next-access delay period (SPND) setting value = 000b (1 QSPCLK cycle)

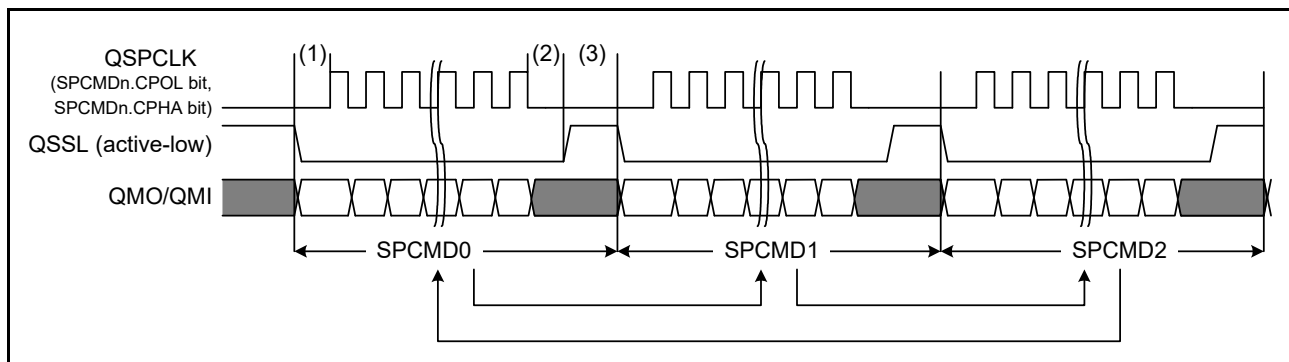


Figure 45.12 Sequence Control Operation (n = 0 to 3)

(d) Burst Transfer

This module can execute burst transfer with the following two methods in single-SPI mode.

One method uses the SPCMDn.SPB[3:0] bits ($n = 0$ to 3) and the SPBMULn register. Setting SPCMDn.SPB[3:0] bits to select 8, 16, or 32 bits and setting the SPBMULn register to select 1 through 4,294,967,296 allows the specified length of data to be continuously transferred for the specified times, where the length is specified by SPCMDn.SPB[3:0] bits and the number of times is specified by the SPBMULn register. However, if the transmit buffer (SPTXB) becomes empty during transfer, or the receive buffer (SPRXB) has no longer a space enough to receive the specified length of data defined by SPCMDn.SPB[3:0] bits, the clock is stopped until transfer is resumed. Figure 45.13 shows a burst transfer example in which SPCMDn.SPB[3:0] bits are set to select 32 bits and the SPBMULn register to select four times thus specifying 128 bits as a total transfer data amount. The following describes operations (1) to (4) in the figure.

- (1) First 32-bit data transfer
- (2) Second 32-bit data transfer
- (3) When the transmit buffer becomes empty or the receive buffer has no longer a space for 32 or more bits, the clock is stopped. Here, the QMO continues outputting the previous value. When data is written to the transmit buffer or an enough space is created in the receive buffer, the clock output is resumed to restart transfer.
- (4) Third and fourth 32-bit data transfer

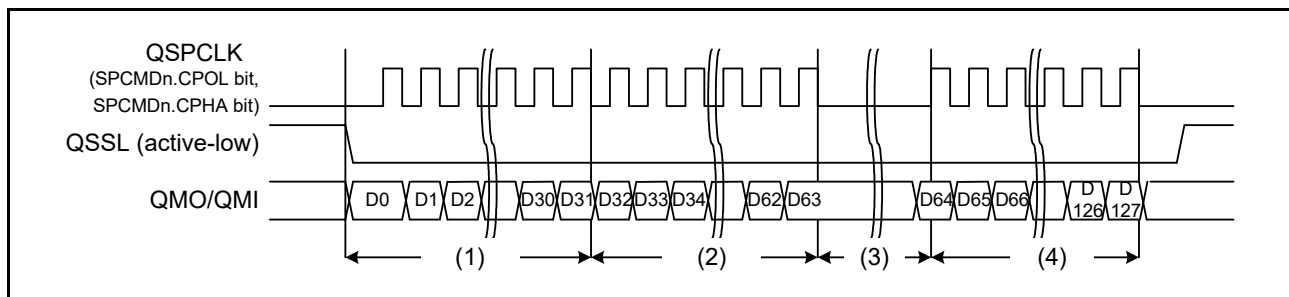


Figure 45.13 Burst Transfer Example in which Total Transfer Data Amount is 128 Bits (Single-SPI Mode)
($n = 0$ to 3)

In the other method, QSSL is kept asserted after a serial transfer is completed until the next serial transfer. Setting the SPCMDn.SSLKP bit ($n = 0$ to 3) allows the QSSL signal to be kept asserted after the transfer corresponding to the SPCMDn register is completed until the next transfer. Figure 45.14 shows a burst transfer example in which the QSSL signal level keeping function is used. The following describes operations (1) to (6) in the figure.

- (1) Clock delay period according to the SPCMD0 register setting. The setting should be made so that the delay period should be at least 1.5 QSPCLK cycles for the first transfer in the burst transfer.
- (2) QSSL negation delay period according to the SPCMD0 register setting. Since the SPCMD0.SSLKP bit is set to 1, QSSL is not negated even after QSSL negation delay period is over. The QSSL negation delay period depends on the SPCMD0.SLNDEN bit setting. When the SPCMD0.SLNDEN bit is 1, the QSSL negation delay period is determined by the SSLND register setting, and the delay period is 0 QSPCLK cycle when the SPCMD0.SLNDEN bit is 0.
- (3) Next-access delay period according to the SPCMD0 register setting. Since the SPCMD0.SSLKP bit is set to 1, QSSL is not negated even during next-access delay period. The next-access delay period depends on the SPCMD0.SPNDEN bit setting. When the SPCMD0.SPNDEN bit is 1, the next-access delay period is determined by the SPND register setting, and the delay period is 0 QSPCLK cycle when the SPCMD0.SPNDEN bit is 0.
- (4) Clock delay period according to the SPCMD1 register setting. The clock delay period depends on the SPCMD1.SCKDEN bit setting. When the SPCMD1.SCKDEN bit is 1, the clock delay period is determined by the SPCKD register setting, and the delay period is 0 QSPCLK cycle when the SPCMD1.SCKDEN bit is 0.
- (5) QSSL negation delay period according to the SPCMD1 register setting. The setting should be made so that the delay period should be at least one QSPCLK cycle for the last transfer in the burst transfer. Since the SPCMD1.SSLKP bit is set to 0, QSSL is negated after QSSL negation delay period is over.
- (6) Next-access delay period according to the SPCMD1 register setting. The setting should be made so that the delay period should be at least one QSPCLK cycle for the last transfer in the burst transfer. Set the SPCMD1.SSLKP bit to 0 to negate QSSL.

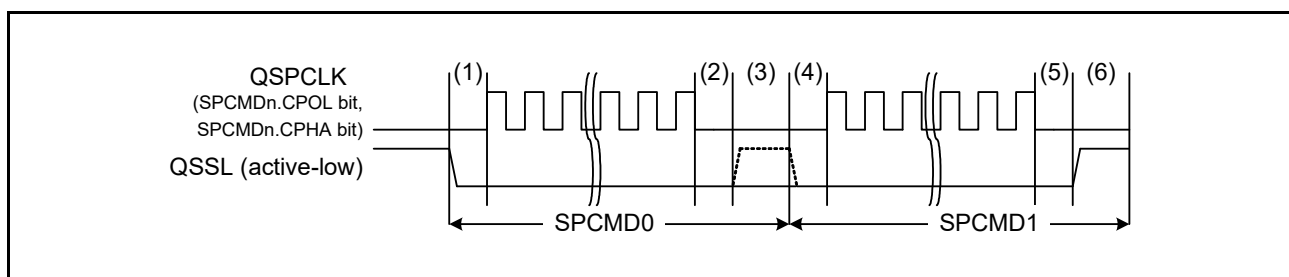


Figure 45.14 Burst Transfer Example in which QSSL Signal Level Keeping Function is Used (Single-SPI Mode) ($n = 0$ to 3)

Note the following when specifying a burst transfer using this method.

Periods (2) to (4) should be inserted without fail when changing the clock polarity through command update.

(e) Initialization Flowchart

Figure 45.15 is a flowchart illustrating an example of initialization in SPI operation when this module is used in single-SPI mode. For a description of how to set up the interrupt controller, DTC, or DMAC, see the descriptions given in the corresponding sections.

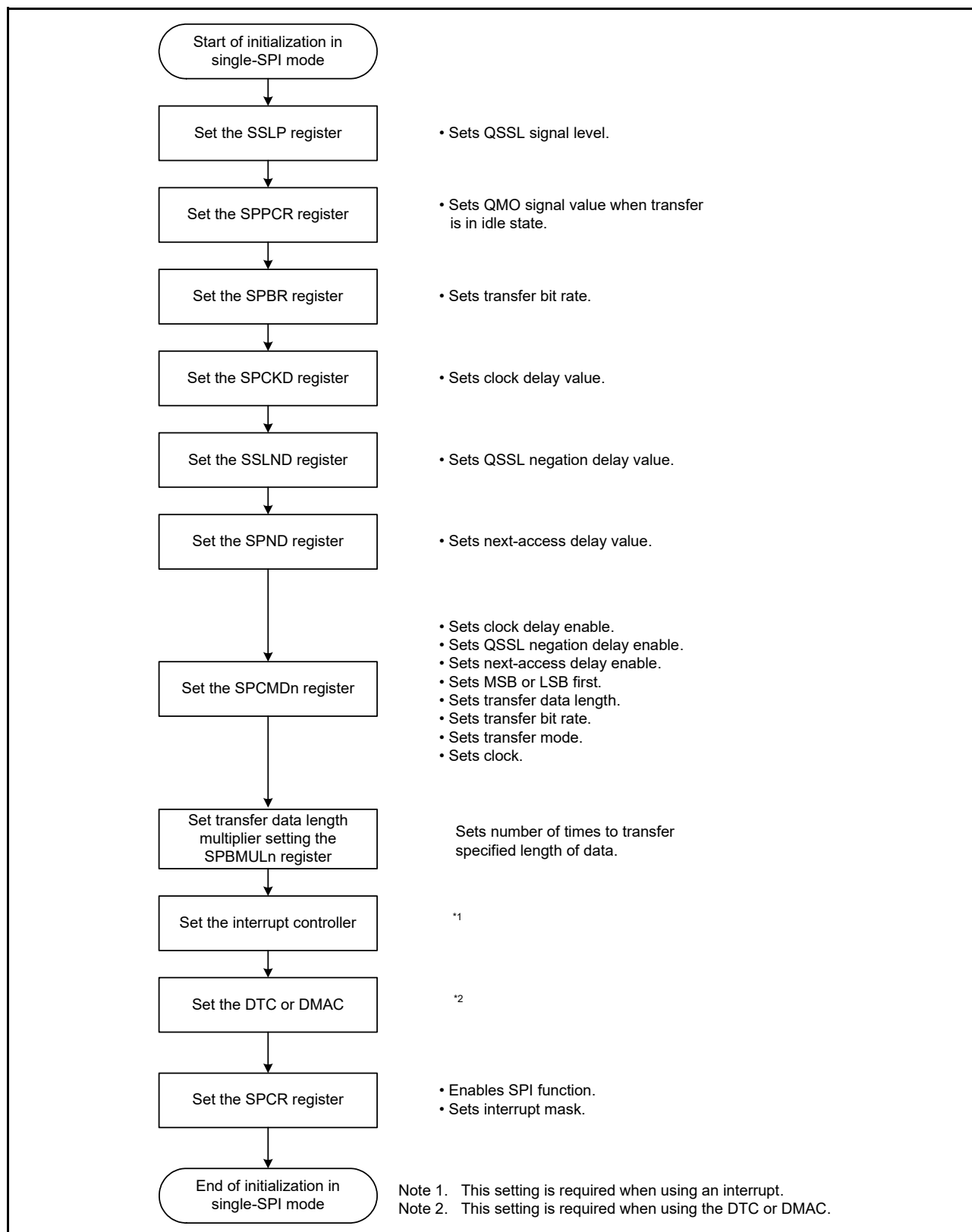


Figure 45.15 Example of Initialization Flowchart in Single-SPI Mode (n = 0 to 3)

(f) Transfer Operation Flowchart

Figure 45.16 is a flowchart illustrating a transfer in SPI operation when this module is used in single-SPI mode. Burst transfer by setting the transfer data length is also executed based on this flowchart.

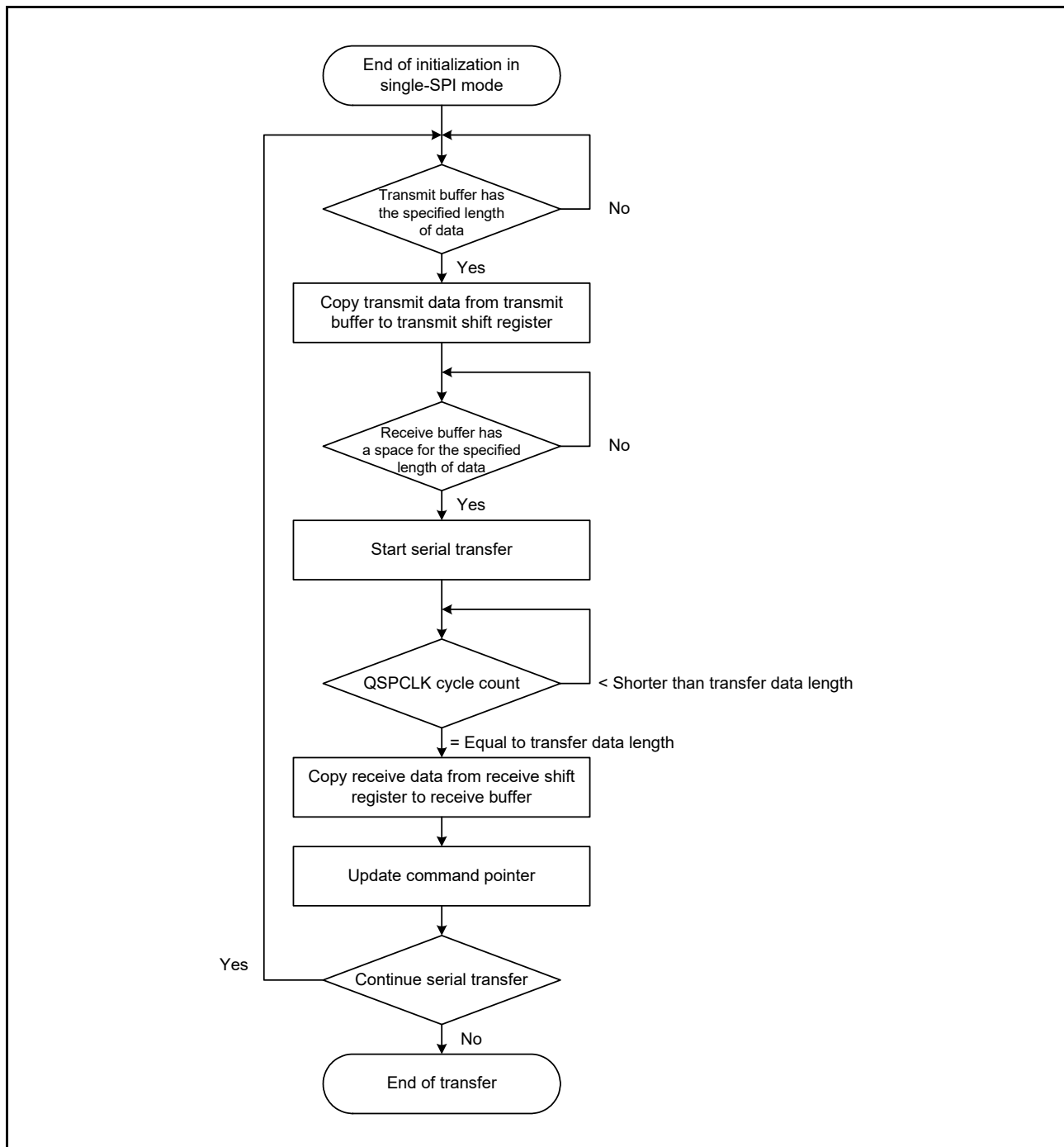


Figure 45.16 Transfer Operation Flowchart in Single-SPI Mode

(2) Dual-SPI/Quad-SPI Mode

(a) Starting Serial Transfer

In dual-/quad-SPI modes, the serial transfer start condition is different depending on the data transfer direction (transmission or reception).

In data transmission, the serial transfer start condition is that there is the specified length of data in the transmit buffer. In data reception, the serial transfer start condition is that there is a space for the specified length of data in the receive buffer.

(b) Terminating Serial Transfer

Irrespective of data transmission or reception, this module terminates the serial transfer after transmitting an QSPCLK edge corresponding to the final sampling timing.

During idle cycles in dual-/quad-SPI modes, the I/O pins are controlled differently depending on whether it is after write or read operation. The I/O pins output either the last output data or the fixed level depending on the register setting after write operation, whereas the I/O pins are driven to the Hi-Z state after read operation. Figure 45.17 shows an example of the pin states after quad-SPI mode access is completed. The following describes operations (1) and (2) in the figure.

- (1) During write operation, QIO0 to QIO3 serve as output pins. Thus, when QSSL is negated upon completion of write operation, QIO0 to QIO3 output different values depending on the value of SPPCR.MOIFE bit. The I/O pins output the level specified by the SPPCR.MOIFV bit when the SPPCR.MOIFE bit is 1, whereas the I/O pins output the last output data when SPPCR.MOIFE bit is 0.
- (2) During read operation, QIO0 to QIO3 serve as input pins. Thus, when QSSL is negated upon completion of read operation, QIO0 to QIO3 are driven to Hi-Z state irrespective of the values of SPPCR.MOIFE and SPPCR.MOIFV bits. In dual-/quad-SPI read operations, receive operation is continued as long as the receive buffer has an enough space for the receive data. To terminate read operation, set the SPCR.SPE bit to 0 after receiving the required length of data, or execute write operation for the last sequence to empty the transmit buffer.

For details on the pin control in dual-/quad-SPI modes, see section 45.3.2, Pin Control.

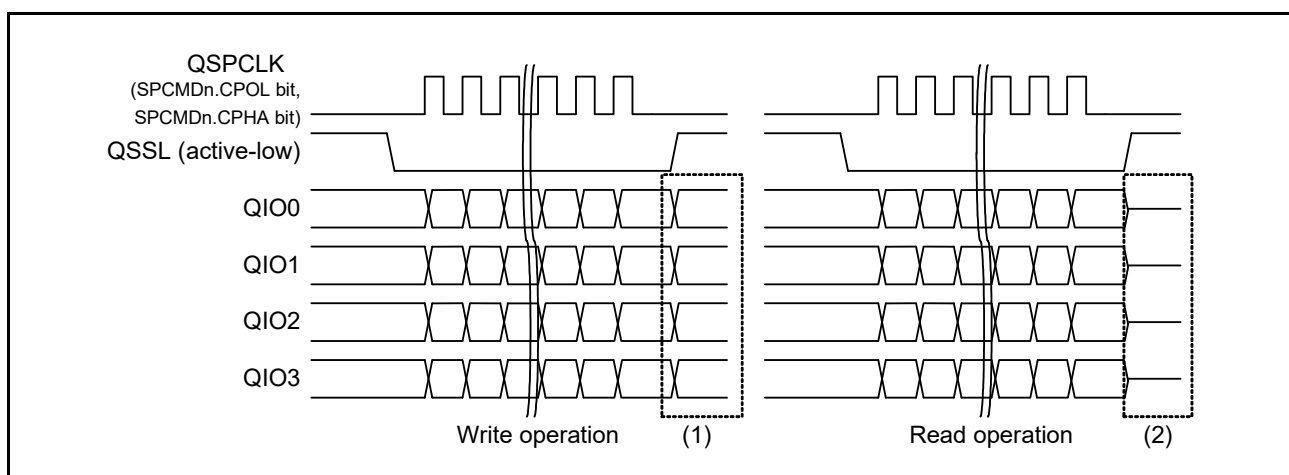


Figure 45.17 Example of Pin States after Quad-SPI Mode Transfer is Completed (n = 0 to 3)

(c) Sequence Control

As with the single-SPI mode, in dual-/quad-SPI modes, according to the sequence length that is assigned to the SPSCR register, this module makes up a sequence comprised of the SPCMDn register ($n = 0$ to 3) and the SPBMULn register. For details on operation, refer to section 45.3.7 (1) (c), Sequence Control.

Dual-/quad-SPI modes only provide operation of a single direction, that is, either transmission or reception for serial transfer. Transmission or reception is set using the SPCMDn.SPRW bit. One of the three operating modes including dual-SPI mode, quad-SPI mode, and single-SPI mode is set using the SPCMDn.SPIMOD[1:0] bits. Combining these bits allow switching single-SPI mode, dual-SPI mode transmission/reception, and quad-SPI mode transmission/reception to control sequence. Figure 45.18 shows an example of sequence configuration with transfer mode switching. For sequence control, use either of the following methods.

- (1) When a sequence is started by write operation and then read operation is performed, terminate the sequence upon completion of read operation. Do not perform write operation again after read operation.

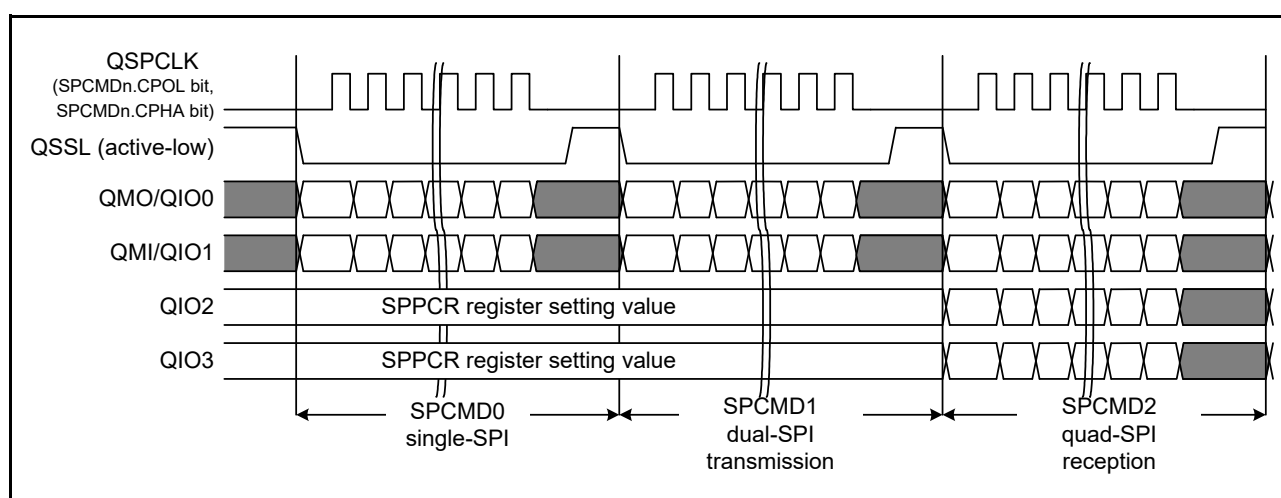


Figure 45.18 Example of Sequence Configuration with Transfer Mode Switching ($n = 0$ to 3)

- (2) Note the following when configuring a sequence in dual-/quad-SPI modes. When all the commands configuring a sequence are dual-/quad-SPI read operations, the sequential operation is continued as long as the receive buffer has an enough space for the receive data. Figure 45.19 and Figure 45.20 show examples of sequence configuration with transfer mode switching.

To terminate read operation, there are the following two methods.

[Termination Method 1]

Set the SPCR.SPE bit to 0 after receiving the required length of data. The receive command is continuously executed, allowing QSPCLK to be output and QSSL to be asserted. However, SPE is forcibly set to 0 when the receive buffer is read as the condition for terminating read operation. At that time, QSSL and QSPCLK are stopped asynchronously.

[Termination Method 2]

After receiving the required length of data, execute write operation in quad-SPI mode for the last sequence and ensure the transmit buffer is empty. Since the transmit command is not started unless transmit data is written, unnecessary QSPCLK is not output and QSSL is not asserted.

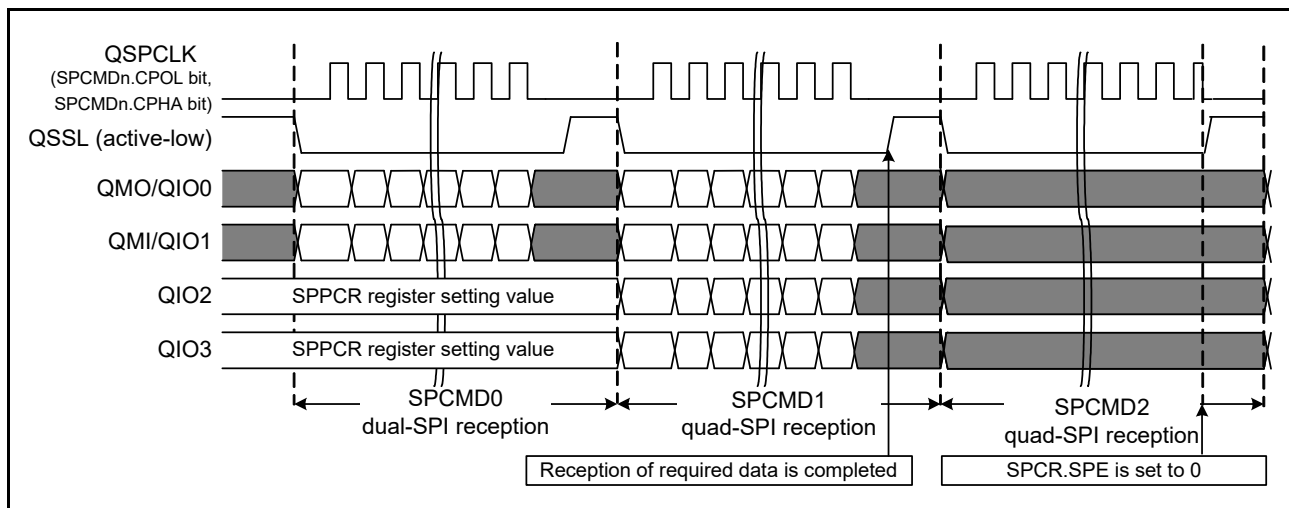


Figure 45.19 Example of Sequence Configuration with Transfer Mode Switching (Termination Method 1 When All the Commands Configuring a Sequence Are Dual-/Quad-SPI Read Operations) (n = 0 to 3)

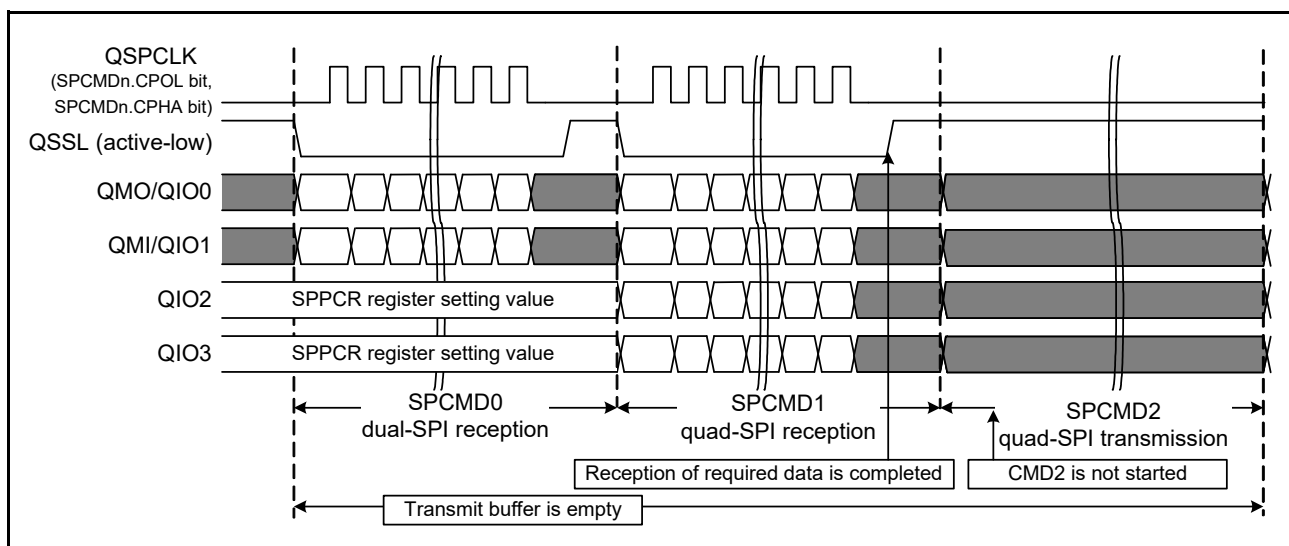


Figure 45.20 Example of Sequence Configuration with Transfer Mode Switching (Termination Method 2 When All the Commands Configuring a Sequence Are Dual-/Quad-SPI Read Operations) (n = 0 to 3)

(d) Burst Transfer

This module can execute burst transfer with the following two methods in dual/quad SPI modes.

One method uses the SPCMDn.SPB[3:0] bits ($n = 0$ to 3) and the SPBMULn register. As with the single-SPI mode, setting the SPCMDn.SPB[3:0] bits to select 8, 16, or 32 bits and setting the SPBMULn register to select 1 through 4,294,967,296 allows the specified length of data to be continuously transferred for the specified times, where the length is specified by the SPCMDn.SPB[3:0] bits and the number of times is specified by the SPBMULn register. However, if the transmit buffer (SPTXB) becomes empty during transfer, or the receive buffer (SPRXB) has no longer a space enough to receive the specified length of data defined by the SPCMDn.SPB[3:0] bits, the clock is stopped until transfer is resumed. This method is effective to transfer a large amount of data in dual-/quad SPI modes. Figure 45.21 shows a burst transfer example in which the SPCMDn.SPB[3:0] bits are set to select 32 bits and the SPBMULn register to select four times thus specifying 128 bits as a total transfer data amount. The following describes operations (1) to (4) in the figure.

- (1) First 32-bit data transfer
- (2) Second 32-bit data transfer
- (3) When the transmit buffer becomes empty or the receive buffer has no longer a space for 32 or more bits, the clock is stopped. Here, when QIO3 to QIO0 serve as output pins, QIO3 to QIO0 continue outputting the previous value. When QIO3 to QIO0 serve as input pins, the inputs to QIO3 to QIO0 depend on the output value of the device to communicate with. When data is written to the transmit buffer or an enough space is created in the receive buffer, the clock output is resumed to restart transfer.
- (4) Third and fourth 32-bit data transfer

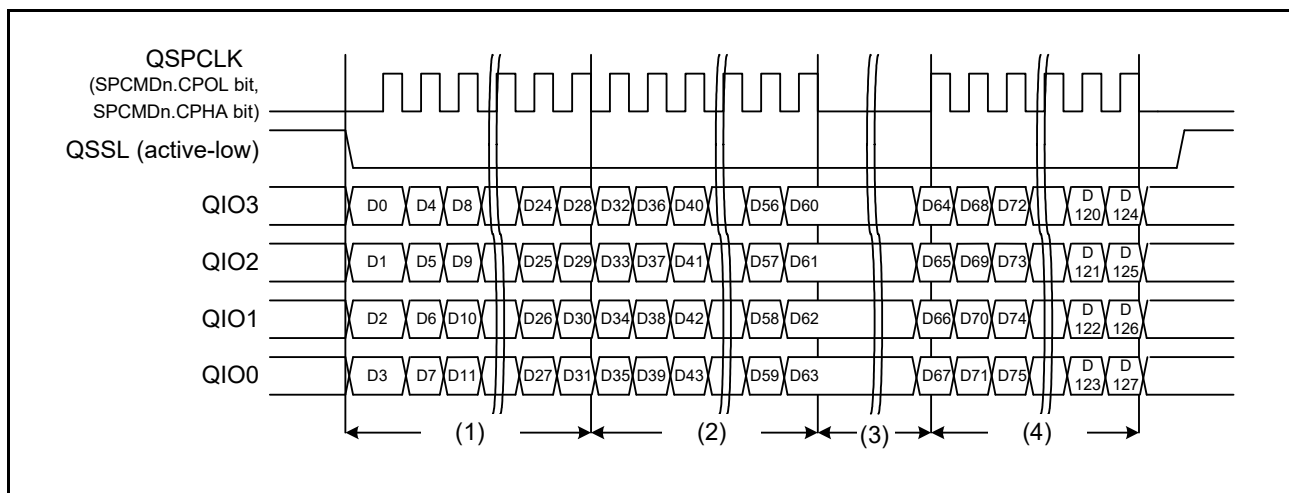


Figure 45.21 Burst Transfer Example in which Total Transfer Data Amount is 128 Bits (Quad-SPI Mode) ($n = 0$ to 3)

The other method uses the QSSL signal level keeping function as in single-SPI mode. Since this method allows switching the SPI transfer modes (single-/dual-/quad-SPI) during a transfer, it is particularly convenient when used with serial flash memory, where command data is written in single-SPI mode and data to be stored in memory is written in quad-SPI mode. Note, however, that at least one delay cycle should be inserted between transfers when switching the SPI transfer modes. Figure 45.22 shows a burst transfer example in which both single-SPI and quad-SPI modes are used. The following describes operations (1) to (6) in the figure.

- (1) Clock delay period according to the SPCMD0 register setting. The setting should be made so that the delay period should be at least 1.5 QSPCLK cycles for the first transfer in the burst transfer.
- (2) QSSL negation delay period according to the SPCMD0 register setting. Since the SPCMD0.SSLKP bit is set to 1, QSSL is not negated even after QSSL negation delay period is over. The QSSL negation delay period depends on the SPCMD0.SLNDEN bit setting. When the SPCMD0.SLNDEN bit is 1, the QSSL negation delay period is determined by the SSLND register setting, and the delay period is 0 QSPCLK cycle when the SPCMD0.SLNDEN bit is 0.
- (3) Next-access delay period according to the SPCMD0 register setting. Since the SPCMD0.SSLKP bit is set to 1, QSSL is not negated even during next-access delay period. The next-access delay period depends on the SPCMD0.SPNDEN bit setting. When the SPCMD0.SPNDEN bit is 1, the next-access delay period is determined by the SPND register setting, and the delay period is 0 QSPCLK cycle when the SPCMD0.SPNDEN bit is 0. Up to this period, the data pin is driven according to the SPCMD0 register setting.
- (4) Clock delay period according to the SPCMD1 register setting. The clock delay period depends on the SPCMD1.SCKDEN bit setting. When the SPCMD1.SCKDEN bit is 1, the clock delay period is determined by the SPCKD register setting, and the delay period is 0 QSPCLK cycle when the SPCMD1.SCKDEN bit is 0.
- (5) QSSL negation delay period according to the SPCMD1 register setting. The setting should be made so that the delay period should be at least one QSPCLK cycle for the last transfer in the burst transfer. Since the SPCMD1.SSLKP bit is set to 0, QSSL is negated after QSSL negation delay period is over.
- (6) Next-access delay period according to the SPCMD1 register setting. The setting should be made so that the delay period should be at least one QSPCLK cycle for the last transfer in the burst transfer. Set the SPCMD1.SSLKP bit to 0 to negate QSSL.

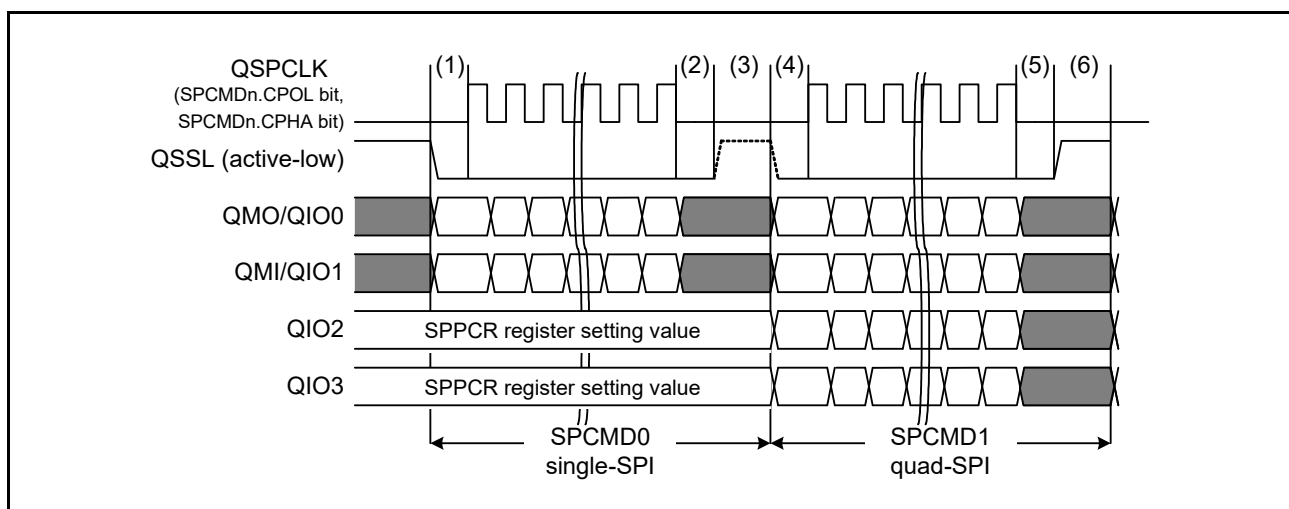


Figure 45.22 Burst Transfer Example in which QSSL Signal Level Keeping Function is Used (Single- and Quad-SPI Modes Used) (n = 0 to 3)

Note the following when specifying a burst transfer using this method.

Periods (2) to (4) should be inserted without fail when changing the clock polarity through command update.

(e) Initialization Flowchart

Figure 45.23 is a flowchart illustrating an example of initialization in SPI operation when this module is used in dual-/quad-SPI mode. For a description of how to set up the interrupt controller, DTC, or DMAC, see the descriptions given in the corresponding sections.

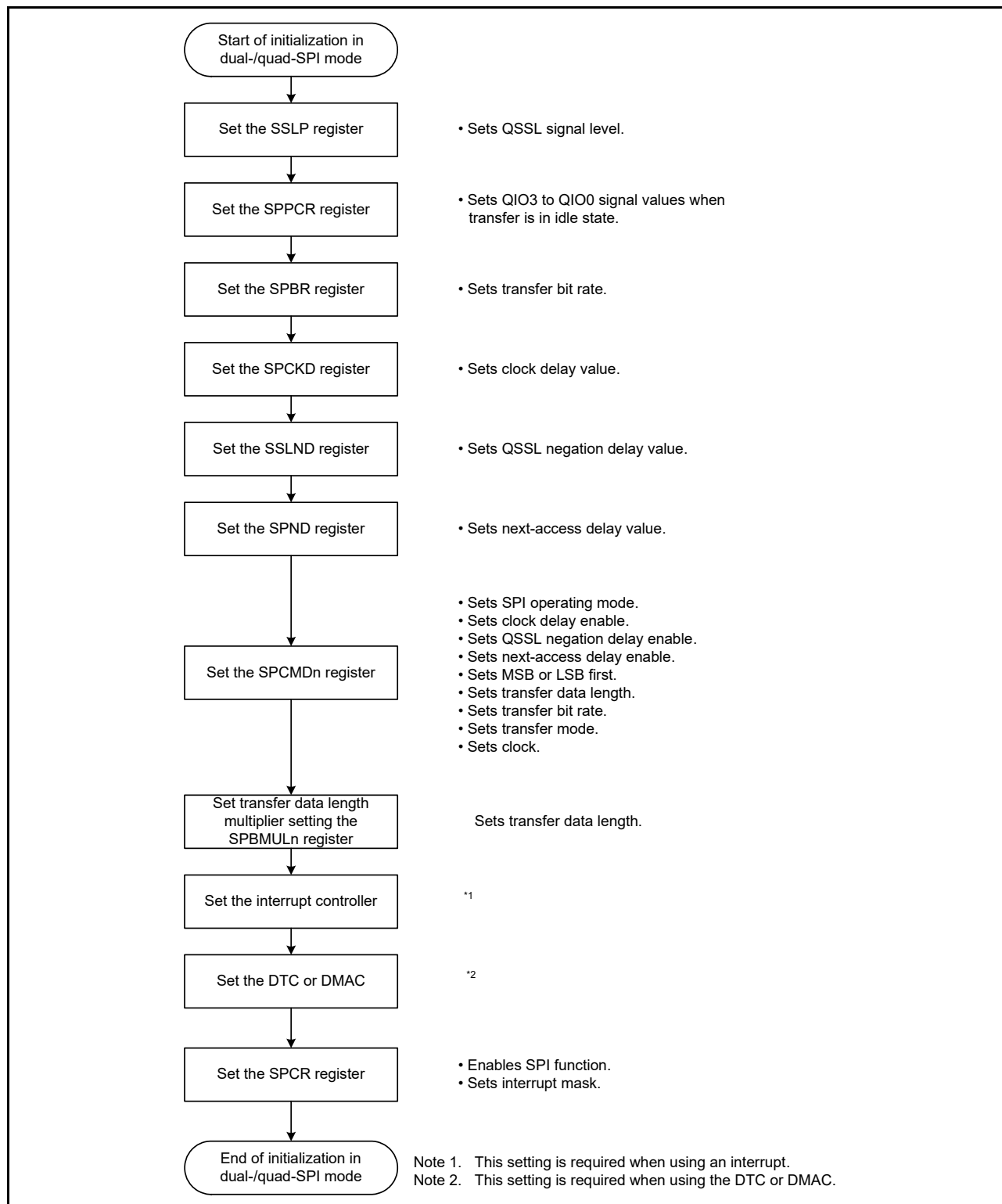


Figure 45.23 Example of Initialization Flowchart in Dual-/Quad-SPI Mode (n = 0 to 3)

(f) Transfer Operation Flowchart

Figure 45.24 shows an operation flowchart (the sequence is started by write operation) and Figure 45.25 shows an operation flowchart (the sequence is configured only with dual-/quad-read operation) in dual-/quad-SPI mode.

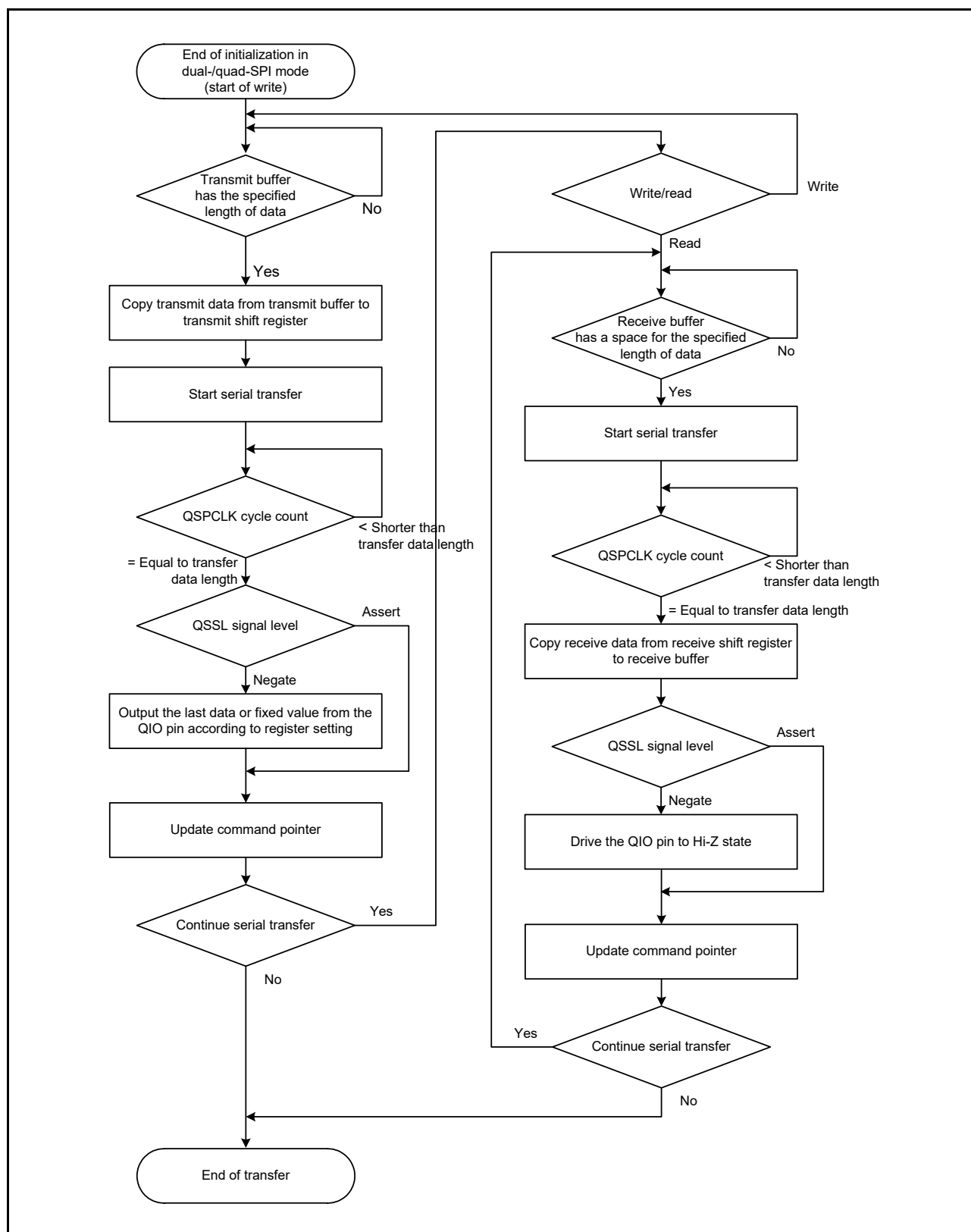


Figure 45.24 Transfer Operation Flowchart in Dual-/Quad-SPI Mode (Sequence is Started by Write Operation)

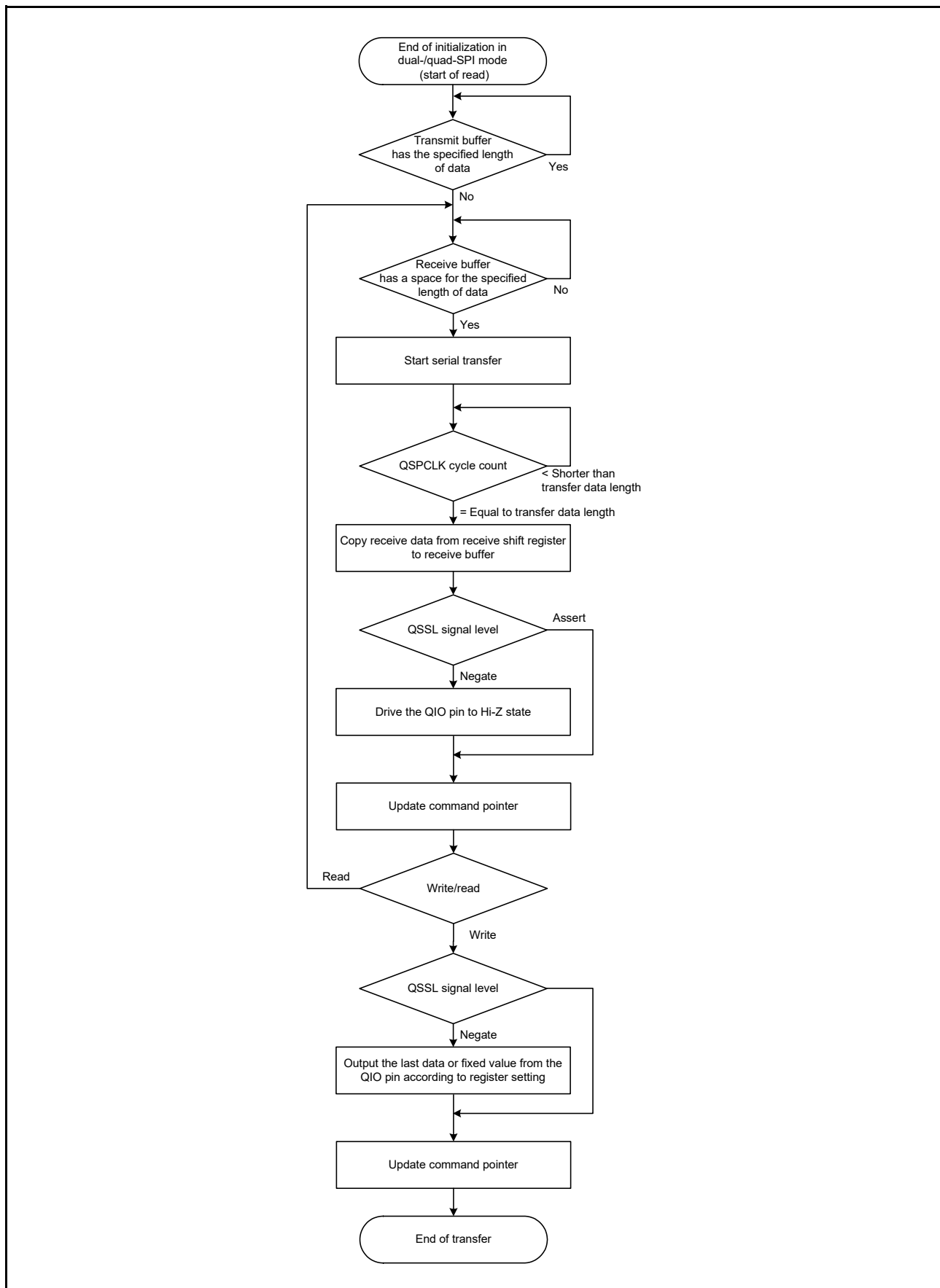


Figure 45.25 Transfer Operation Flowchart in Dual-/Quad-SPI Mode
(Sequence is Configured only with Dual-/Quad Read Operations)

45.3.8 Interrupt Sources

This module has interrupt sources of receive buffer full, transmit buffer empty and QSSL negation. In addition, the DTC or DMAC can be activated by the receive buffer full or transmit buffer empty interrupt for data transfer.

Table 45.10 shows the interrupt sources. When any of the interrupt conditions in Table 45.10 is met, an interrupt is generated. The interrupt sources should be cleared with data transfer by the CPU, DTC, or DMAC.

Table 45.10 Interrupt Sources

Name	Interrupt Source	Interrupt Condition
SPRI	Receive buffer full	SPSR.SPRFF flag = 1 and SPCR.SPRIE bit = 1
SPTI	Transmit buffer empty	SPSR.SPTEF flag = 1 and SPCR.SPTIE bit = 1
QSPSLI	QSSL negation	SPSR.SPSSLF flag = 1 and SPCR.SPSSLIE bit = 1

If the conditions for a transmit buffer empty interrupt (SPTI) and receive buffer full interrupt (SPRI) are satisfied while the ICU.IRn.IR flag is 1, the interrupt request is not output to the ICU, but is retained internally. When the ICU.IRn.IR flag is set to 0, the retained interrupt request is output to the ICU.

For the ICU.IRn.IR flag, refer to section 15, Interrupt Controller (ICUD).

If the interrupt source (transmit buffer empty or receive buffer full) for the transfer buffer empty interrupt or receive buffer full interrupt is not cleared upon completion of transfer, the interrupt request generation conditions are as follows.

- DTC or DMAC transfer: When the interrupt condition is satisfied after the completion of access with the last DTC or DMAC transfer.
- CPU interrupt: When the interrupt condition is satisfied after the status flag has been cleared by register access.

Note that while the DTC or DMAC is in use, access to the flags by register reading or writing is prohibited.

45.3.9 Loopback Mode

This module provides loopback mode for testing. Writing 1 to the SPPCR.SPLP bit enables loopback mode. In loopback mode, this module connects the outputs from the transmit shift register to the inputs to the receive shift register instead.

Figure 45.26 shows a schematic internal connection in loopback mode.

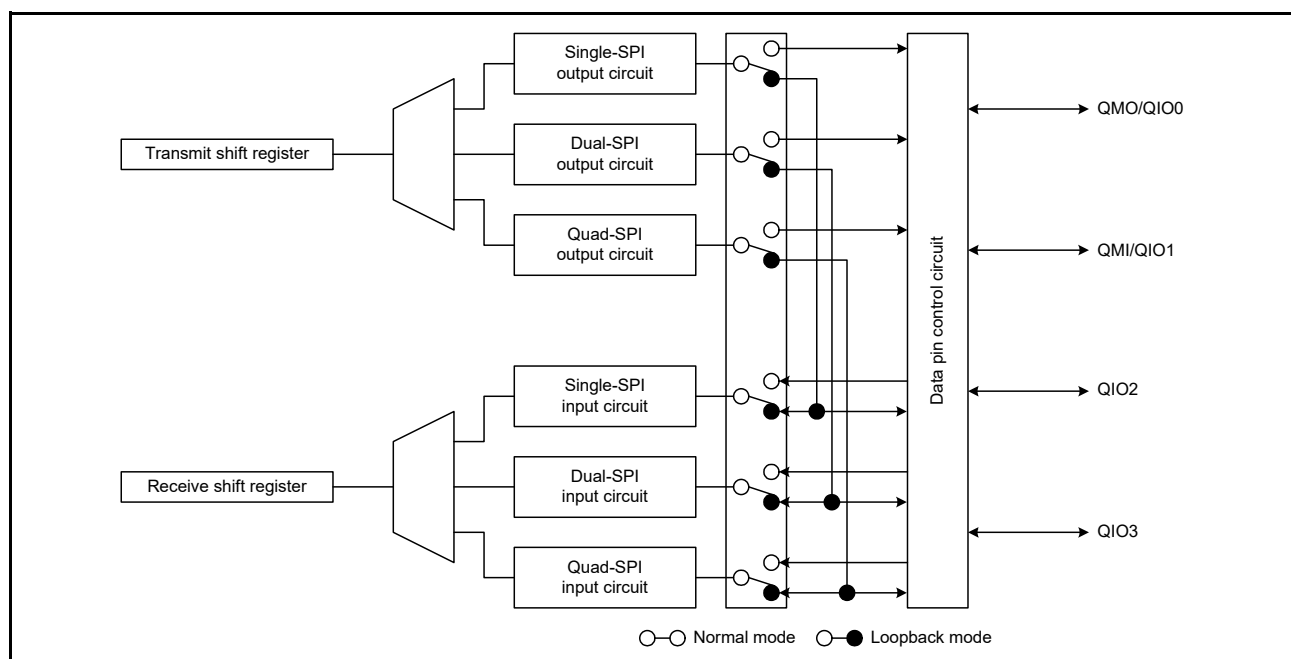


Figure 45.26 Schematic Internal Connection in Loopback Mode

45.4 Usage Notes

45.4.1 Notes on Starting Transfer

When the ICU.IRn.IR flag for the DTC or DMAC is 1 at the time transfer is to start, follow the procedure below to clear interrupt requests before enabling operations (by setting the SPCR.SPE bit to 1).

- (1) Confirm that transfer has stopped (i.e. that the SPCR.SPE bit is 0).
- (2) Set the relevant interrupt enable bit (the SPCR.SPTIE or SPCR.SPRIE bit) to 0.
- (3) Read the relevant interrupt enable bit (the SPCR.SPTIE or SPCR.SPRIE bit) and confirm that its value is 0.
- (4) Set the ICU.IRn.IR flag to 0.

45.4.2 Module Stop Function Setting

Operation of the QSPI can be disabled or enabled by module stop control register C (MSTPCRC). The initial setting is for operation of the QSPI to be stopped. Register access is enabled by releasing the module stop state. For details, refer to section 11, Low Power Consumption.

45.4.3 Notes on Using the Serial Flash Memory

When using the serial flash memory in dual- or quad-SPI operating mode, set the SPCMDn.CPOL and CPHA bits ($n = 0$ to 3) to 1 and select SPI mode 3. SPI mode 0 to 2 cannot be used. In addition, set the SPCMDn.SPNDEN, SLNDEN, and SCKDEN bits to 1 to secure delay period.